

“I Know My Method, but Not How to Show It”: Human–AI Collaborative Progressive Intent Specification for Methodology Figure Authoring

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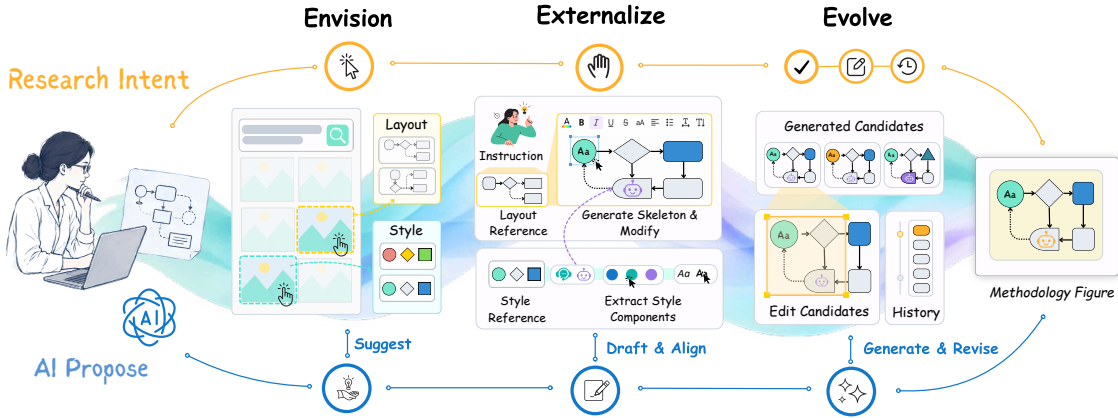


Fig. 1. Overview of the HiFIGURE authoring workflow. In **Envision**, authors explore and select layout and style references with AI suggestions. In **Externalize**, they generate and modify an editable figure skeleton and extract visual components from style references. In **Evolve**, they generate, inspect, and revise figure candidates, with access to revision history, to develop the final methodology figure. The orange and blue paths indicate Human action and AI support, respectively.

Researchers may understand their methods yet struggle to represent them visually. Existing prompt-based generative artificial intelligence (AI) systems reduce drawing effort but offer limited support for forming and refining visual intent. To examine how researchers progressively specify visual intent during methodology figure authoring, we conducted a formative study with five researchers. We conceptualize the authoring process as *Progressive Intent Specification*, in which researchers iteratively envision, externalize, and evolve a method’s visual representation. Building on these findings, we present HiFIGURE, a human–AI collaborative system supporting these three interdependent activities. In a within-subjects study with 30 participants, participants reported significantly greater creativity support, usability, ownership, and agency with HiFIGURE than with a prompt-only generative AI baseline. The resulting methodology figures received higher quality ratings from both participant self-perception and expert evaluations. Analysis of interaction logs and interviews further shows how participants revisit earlier decisions and refine provisional representations throughout the authoring.

CCS Concepts: • **Human-centered computing** → *Interactive systems and tools*.

Additional Key Words and Phrases: methodology figures, human–AI collaboration, generative AI, progressive intent specification, scientific visualization

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1 Introduction

Methodology figures are a central form of visual explanation in computing research [56]. Creating a methodology figure can be understood as a process of *visualizing a method*, translating a researcher's understanding of the method into a visual representation that communicates its components, relationships, and progression. However, this translation is inherently underdetermined. A method does not merely stay in one single visual form. Depending on the researcher's intent, the same method can be visually translated into a linear pipeline, a feedback loop, or a set of parallel stages, each emphasizing different relationships and guiding readers through the method differently [29, 54]. A key challenge in this translation is therefore *intent specification*. Researchers may know their method, but may not yet have a fully specified idea of how it should be visually represented. Rather than being determined upfront, decisions about what to include, how to organize it, and what to emphasize can emerge and evolve during authoring [18]. Prior work has shown that visual representations do not simply record existing ideas [40]; they can also serve as provisional representations through which people discover and revise how a problem should be represented [71, 81, 85]. In this sense, visual intent is progressively specified through the creation, inspection, and revision of provisional representations. We refer to this process as **Progressive Intent Specification**.

In conventional authoring workflows, researchers progressively develop their visual intent by searching for references, sketching alternative representations, and iteratively refining visual details as their ideas evolve [3, 33, 54]. However, these workflows require researchers to manually implement and coordinate emerging visual decisions, often across multiple tools. Recent generative artificial intelligence (AI) systems, such as AutoFigure [102], PaperBanana [101], and their interactive extensions [48, 94], substantially reduce this execution burden by supporting automatic generation and iterative refinement. However, reducing the burden of execution does not necessarily address the challenge of intent specification. These prompt-based prompting systems leave two challenges insufficiently addressed. Researchers may lack the awareness and knowledge needed to progressively turn an initial idea into a specific intent. This reflects the *gulf of envisioning*, where users must first develop an actionable intent before they can effectively express it to an AI system [77]. Moreover, prompt-based systems provide limited visual support for specifying and revising visual choices while the researchers' intent is still evolving.

Existing interactive design systems suggest one way to bridge this gap by externalizing emerging intent into representations that users can inspect and manipulate [24]. For example, Brickify [74] supports visual specification through manipulable design tokens, while ToMigo [31] uses editable concept graphs to help users develop and revise design intent. These systems demonstrate the value of making partially formed intent explicit during authoring. However, authoring the methodology figure introduces an additional challenge because visual decisions must remain tied to the scientific meaning they are intended to communicate. Researchers may change how a method is visually organized while still needing to inspect whether its components, relationships, and information flow are represented faithfully. Existing approaches provide limited support for jointly representing the scientific structure of a method. Researchers, therefore, lack a clear way to check whether visual changes still reflect the intended scientific meaning. This raises a key HCI challenge: *How can human-AI collaboration support researchers in progressively specifying methodology figures while keeping scientific structure inspectable and visual decisions revisable?*

To understand how researchers approach the progressive intent specification and what visual support they need during methodology figure authoring, we conducted a formative study with five researchers who regularly create scientific figures. Our findings characterize the methodology figure authoring process as a **Progressive Intent Specification** workflow across three interdependent activities of authoring: **Envision** frames a possible visual direction by drawing on references; **Externalize** makes emerging scientific and visual decisions inspectable; and **Evolve** uses provisional outcomes to reconsider and refine those decisions.

Guided by this workflow, we developed **HiFigure**, a Human-AI Collaborative System for methodology Figure authoring from simply prompting generative AI systems toward *Progressive Intent Specification*. HiFIGURE supports authors in developing the representation of their methodology figures, while keeping scientific and visual decisions inspectable and revisable throughout the authoring process. During Envision, HiFIGURE supports reference exploration and allows authors to distinguish the roles of selected references in layout and style. Externalize turns scientific content into an editable figure skeleton and allows visual elements from references to be connected to specific scientific components. Evolve keeps generated candidates linked to the specification that produced them and supports local revision, comparison, and return to earlier specification states. It shifts the methodology figure authoring away from repeatedly describing a desired output and toward developing the representation itself through interaction.

We evaluated HiFIGURE in a within-subject study with 30 participants, comparing it with a prompt-based AI workflow. Participants selected either NLP or CV based on their familiarity and completed two methodology figure authoring tasks within that domain, one with each system. Participants reported significantly greater creativity support, usability, sense of ownership, and agency with HiFIGURE, and rated the resulting figures significantly higher in readability and overall quality. Two experts also independently evaluated all 60 figures generated during the study. Both experts assigned significantly higher ratings to figures created with HiFIGURE in overall quality and across all eight evaluation dimensions. These findings suggest that HiFIGURE’s three-stage Progressive Intent Specification workflow can support authors’ involvement and control by allowing them to progressively externalize and refine their intent through provisional representations.

This work makes three contributions:

- We derive a *Progressive Intent Specification* workflow for methodology figure authoring from a formative study. The workflow comprises three recurring and interdependent activities: *Envision*, *Externalize*, and *Evolve*.
- We present HiFIGURE, a human-AI collaborative system that supports *Progressive Intent Specification* workflow by externalizing evolving visual intent into inspectable and revisable representations.
- We evaluate HiFIGURE in a within-subject study with 30 participants, showing higher perceived creativity support, usability, ownership, agency, readability, and overall figure quality than a prompt-based AI workflow.

2 Related Work

2.1 Intent Formation and Specification in Visual Authoring

In visual authoring, intent can develop through articulating goals, exploring sketches and reference images, and evaluating design alternatives [19, 24, 53, 80]. These activities allow authors to reconsider both what they want to communicate and how it should be represented, as emerging designs reveal possibilities that were not apparent at the outset [18, 80].

Articulating and developing intent. *Intent specification* concerns making goals, requirements, and preferences explicit in representations that can guide a system [24, 44, 86]. Earlier work by Leveson used intent specifications

to organize system and software specifications around human understanding and problem solving [44]. In visual authoring, Brickify uses *design intent specification* to describe how visual elements are expressed and how they should be composed [74]. In recent AI research, Semantic Commit describes intent specifications as evolving, human-readable records of user goals, preferences, and requirements that guide AI behavior and remain available for inspection and revision [86]. The specified intent can itself develop during interaction [24, 80]. Subramonyam et al. describe the *gulf of envisioning*, highlighting the challenge of translating broader goals into actionable intent and instructions for an LLM [77]. Studies of prompt design also document difficulties in communicating requirements that produce the intended behavior [24, 49, 98]. Design research shows how exploring possible solutions can change designers’ understanding of the problem itself [18, 80]. These findings motivate the supporting authors to work out their visual intent as they express and evaluate it.

External representations and reference exploration. During visual authoring, external representations are visible forms of ideas and information, such as sketches, diagrams, and reference images [40, 53]. Authors can inspect and change these representations as they decide what to communicate. Studies of external representations help explain how such interaction contributes to the development of design ideas [25, 40]. Suwa and Tversky describe how architects interpret their own drawings and use what they notice to develop their ideas [81]. Tversky further explains how ambiguity in sketches can allow new interpretations and design possibilities [85]. For data structure diagrams, Hayatpur et al. describe how visual form, simplification, and annotation affect what information is communicated [29]. What authors notice in a representation can change both the idea and how they choose to show it [25, 80]. Reference figures offer another form of external representation, giving designers examples to consider when developing a visual direction [33, 87]. Interviews with visualization designers describe how they interpret inspiration from diverse sources and relate it to their own work [3]. Several systems support this reference exploration through interaction with reference material [41, 42, 75]. SemanticCollage uses semantic labels to help users organize and explore images on mood boards [42]. GenQuery lets users generate and edit images during visual search, helping them explore and express what they are looking for [75]. CreativeConnect supports graphic design ideation by letting users select and recombine concepts from reference images [15]. Creating with references in these ways helps designers decide more precisely what they want to create [41, 53].

Drafts and iterative clarification. Drafts give authors concrete alternatives to inspect and compare [19, 28]. Dow et al. found that developing multiple prototypes in parallel led to more varied designs and improved outcomes in graphic advertising tasks [19]. Side Views supports such experimentation through persistent preview of alternative command outcomes before changes are committed [83]. Generative systems provide further ways to explore alternatives: Inkspire lets users move between sketches and AI-generated product concepts [46], while Luminate organizes LLM-generated alternatives along design dimensions [78]. Promptify combines prompt suggestions with an interface for organizing generated images and requesting refinements, allowing users to develop subsequent prompts from their responses to earlier results [12]. In the formative study that motivated PaperBanana-Interact, nine of fourteen participants reported that seeing an initial diagram shaped or changed how they wanted it presented [94]. Viewing a draft can therefore help authors clarify the intent that guides subsequent revisions [24, 28]. Across these studies, sketches, references, and drafts help authors form and reconsider intent in their work. This perspective informs our account of *Progressive Intent Specification*, in which specifications are progressively articulated and remain open to revision during authoring, rather than being fully determined in advance.

2.2 Methodology Figure Authoring and AI Support

Methodology figures communicate how a proposed method works by selecting and organizing its components, operations, and relationships [32, 57]. Their construction involves decisions about both scientific content and visual representation: which parts of a method deserve attention, what level of detail readers need, and how the figure should connect to the explanation in the paper [55, 69]. Research on scientific diagramming examines these communicative decisions, while authoring tools provide computational support for translating scientific descriptions into graphical form.

Scientific content and visual communication. Studies of diagramming practices show that authors actively develop the representations through which scientific concepts become understandable [11, 54]. Through interviews with 18 domain experts, Ma’ayan et al. identify practices of finding, creating, adapting, storing, and reusing visual representations [54]. Authors draw on established representations where available and develop new ones when existing conventions do not adequately express their concepts. Their work highlights the translation between domain knowledge and graphical form as a substantive part of diagram authoring. For neural network architecture diagrams, Marshall et al. identify several reader purposes, including understanding system operation, recognizing a paper’s contribution, and examining dependencies between components [56]. Their findings also reveal differences in the level of detail readers expect and the meanings they assign to visual conventions. The resulting guidance framework addresses appropriate content, visual clarity, and consistency with readers’ expectations. These findings suggest that a methodology figure’s usefulness depends on how its selected content and graphical organization support a particular explanation [30, 32, 69]. A comprehensive depiction of implementation details may serve a different purpose from an overview that foregrounds the method’s main contribution [55, 56].

Diagram construction from scientific structure. Tools can support this translation by deriving diagrams from explicit descriptions of scientific objects and their relationships [4, 23, 43]. Net2Vis generates publication-oriented architecture visualizations from Keras models using a visual grammar for layer encoding, network layout, aggregation, and legend generation [5]. Its aggregation mechanisms allow repeated layer sequences to be represented as higher-level blocks, connecting the underlying computational architecture with the abstraction level of the published figure. More general diagramming frameworks separate conceptual structure from its graphical realization [34, 96]. Penrose specifies mathematical objects and relationships independently of the rules that map them to visual elements, then uses constraints and optimization to produce diagrams [96]. Different mappings can therefore depict the same mathematical content through different visual representations. Bluefish supports the construction of diagrams through composable declarative relations among graphical elements, accommodating structures across domains such as mathematics, physics, and computer science [65]. These approaches encode structural information that would otherwise be maintained through manual drawing [23, 34]. These approaches provide a foundation for translating explicit scientific structure into visual organization, while methodology figure authoring further requires authors to decide what explanatory content to include and how it should be abstracted.

AI-assisted scientific figure generation. General-purpose models such as GPT-Image-2 [60], Nano-Banana-Pro [66], Qwen-Image-2.0 [99], and FLUX.2 [10] support instruction following, text rendering, and complex spatial composition. Methodology figures additionally require faithful representations of scientific components, operations, and relationships. DiagrammerGPT [97] and DiagramAgent [90] use intermediate plans to organize diagram content and layout, followed by image rendering or code generation and checking. Scientific illustration systems extend these

workflows to research content through coordinated generation and iterative refinement [26, 50, 73, 79, 89, 101, 102], with PaperBanana incorporating retrieved reference figures. SciFig [36] produces editable XML methodology figures through staged planning, layout synthesis, rendering, and refinement, while Crafter [100] coordinates planning, rendering, and verification through a shared specification across academic figures, posters, and infographics. Graphics program synthesis provides structured representations of generated outputs through TikZ or SVG programs, enabling explicit control over graphical elements [7, 8, 50, 68, 91, 95]. DeTikZify [9] translates sketches and existing figures into TikZ programs, while Draw with Thought [17] reconstructs raster diagrams as mxGraph XML. AutoFigure-Edit instead generates SVG illustrations from scientific text with reference-guided styling [48]. These approaches offer multiple routes from research materials to scientific figures, combining content interpretation, visual organization, and structured rendering.

2.3 Interactive Specification in Human–AI Visual Authoring

A significant challenge of human–AI visual authoring lies in using natural language to specify spatial relationships, visual attributes, and the intended scope of changes [47, 64, 70, 74]. Interactive authoring systems address these difficulties by making aspects of the desired result available as objects that users can select, arrange, and revise.

Making design choices explicit. Representing individual choices separately allows users to inspect and adjust particular aspects of a developing design [24, 51, 63]. Brickify turns subjects, styles, and colors from reference images into reusable design tokens [74]. Users arrange subject tokens to express spatial relationships and link style or color tokens to indicate their intended influence and scope. InkIdeator makes cultural symbols, emotions, composition, and style available as keywords extracted from Chinese paintings, enabling designers to select and combine these concepts to generate visual examples [92]. ToMigo represents choices about purpose, content, and style in editable design concept graphs inferred from textual briefs and reference images [31]. Users can examine relationships between these choices, edit the graph, and respond to reflective questions as they develop their design. These systems make selected aspects of visual intent explicit enough to guide generation while remaining available for further consideration.

Combining textual and graphical specification. Text and graphical interaction offer complementary ways to communicate a desired result or revision [64, 70, 84]. SketchFlex combines rough region sketches with textual descriptions, allowing users to indicate where objects should appear and what those objects should depict [47]. PromptPaint lets users mix prompts and apply them to different canvas regions and at different points in the generation process [16]. PromptCharm visualizes model attention to help users inspect associations between prompt words and the regions of the generated image, and supports attention adjustment and region-based inpainting [88]. Across these approaches, text describes desired content or changes, while graphical actions communicate placement, scope, or relative influence. This combination is relevant to methodology figures, where a revision may concern both a specific visual element and the scientific relationship it represents.

Supporting revision across iterations. As authors inspect results, they may need to update an earlier specification, revise the current image, or return to another design alternative [2, 21, 84]. Systems support these activities through intermediate representations, retained requirements, and accessible histories. Beyond static images, SimStep uses task-level abstractions to help educators incrementally specify, inspect, and debug AI-generated educational simulations [39]. Its approach illustrates how representations organized around domain concepts can provide opportunities to review and revise the choices guiding generation. In scientific diagram authoring, PaperBanana-Interact retains information

from earlier interactions and evaluates revisions against current feedback, previous requirements, and the original scientific context [94]. DesignTrace records visual and semantic design development through command histories, state tracking, and an interactive branching structure [62]. Designers can compare alternatives, revisit earlier decisions, and continue exploring from previous versions. These approaches connect revision of an output with access to the decisions and alternatives developed during authoring. Our work builds on these approaches by focusing on interactive specification in methodology figure authoring, where researchers must coordinate decisions about scientific components and relationships with their visual presentation. Through *Progressive Intent Specification*, HiFIGURE supports researchers in making these choices explicit and revising them as they inspect and refine generated figures.

3 Formative Study

3.1 Methods

Participants: We recruited five computer science researchers for our formative study: one industry researcher (**FP1**), one postdoctoral researcher (**FP2**), and three Ph.D. students (**FP3**, **FP4**, **FP5**). Their research experience ranged from three to five years, and each had created or revised at least six scientific figures in the preceding year. Four participants incorporated AI tools into their current figure-authoring workflows, while **FP2** primarily relied on a manual workflow.

Procedure: We conducted five semi-structured interviews via online conferencing software, each lasting approximately 40 minutes. After providing informed consent, participants described their research backgrounds and experiences with creating scientific figures. The interviews explored how they developed an initial idea or method into a complete figure, searched for and used visual references, integrated AI image-generation tools into their workflows, and decided which tasks to delegate to AI or control manually. We also asked what they envisioned as the ideal tool for creating scientific figures. These questions served as guiding prompts, with follow-up questions used to explore participants' experiences and responses. We conducted all interviews in the participants' native language.

Prototype Demonstration: To reduce potential priming from exposure to the system, we demonstrated the prototype only after the interview questions were completed. Participants compared the prototype with their workflows, discussed its strengths and limitations, suggested improvements, and considered whether they would use it. The prototype used in this study is shown in Appendix A.

Data Analysis: Our corpus comprised five automatically transcribed interviews. We conducted an inductive thematic analysis of participants' authoring practices, workflows, and needs, separating accounts of existing practices from responses to the design probe. We open-coded the transcripts, compared codes across participants, and grouped recurring patterns into themes while retaining divergent cases. Given the small sample drawn from computer science, we aimed to identify design-relevant patterns rather than claim thematic saturation or population-level prevalence.

3.2 Formative Findings

Our findings identified three recurring and interdependent activities through researchers' **Progressive Intent Specification Workflow**: *Envision*, *Externalize*, and *Evolve*.

3.2.1 Envision: Form visual direction from partial references. Reference retrieval generally occurred before formal drawing, which helped researchers establish an initial visual direction. **FP4** said that “*searching happened mostly before drawing*” and the reference pictures had already been “*thought through*” and “*found*”. **FP1**, **FP2**, **FP3**, and **FP5** similarly stated, “*I look at them before I start drawing*,” although they sometimes made a rough module sketch before searching the reference example. We therefore treat the reference retrieval as part of visual authoring, which helped

researchers envision possible layouts and visual languages before formal drawing. Researchers rarely adopted one reference wholesale. Instead, they extracted different visual attributes from multiple sources. **FP3** explained, “For layout, I might use the whole figure; for color or typography, I only refer to part of it.” **FP4** similarly said, “I use part of each figure,” while **FP1** saved examples for “color palettes, themes, and skeletons.” **FP5** prioritized references whose information was direct and whose meaning could be understood without excessive complexity. References were therefore evaluated not only by visual similarity but also by whether they supported the intended scientific communication.

Existing reference-management practices remained lightweight. **FP1** stored references in folders or photo albums, **FP3** placed useful figures beside the Draw.io canvas, and **FP4** passed multiple screenshots directly to AI. Reactions to the design probe also differed: **FP5** considered a dedicated reference pocket unnecessary, while **FP2** thought provenance might make the system “somewhat more trustworthy.”

Finding 1. Researchers envisioned an initial visual direction by selectively extracting attributes from multiple references. However, they differed in how to organize the selected references during authoring.

3.2.2 Externalize: Make evolving visual intent inspectable. Researchers rarely began with a complete specification of the final figure. They generally understood the scientific topic and major components, but the detailed structure became clearer through drawing, comparison, and revision. **FP3** described this process directly: “I form it while drawing.” **FP1** similarly observed, “The more I discussed it with AI, the clearer it became.” **FP4** wrote the method’s inputs, outputs, architecture, and symbols into a prompt, then reconsidered the representation by inspecting successive outputs.

Researchers’ narratives show that the main difficulty was not operating the drawing software but deciding how to represent scientific content. **FP2** explained, “The difficult part is not the drawing itself. It is how to show the modules of a model in the figure—which parts need to be included and what should be shown.” **FP1** identified “layout and reading order” as particularly difficult because AI might include the requested content without considering “how a person will understand it.” **FP4** likewise expected AI to misunderstand “scientific accuracy” and “the relationships among components.”

Intermediate artifacts helped expose these unresolved decisions. **FP3** first sketched the constituent modules and later asked AI to critique an existing version. **FP5** followed a more extended sequence: clarify the content, request an AI draft, conduct several rounds of dialogue, reconstruct the large modules in Draw.io, add detailed content and domain-specific icons, and finally adjust alignment and color. In each case, the intermediate artifact helped to reveal missing content, unclear relationships, or unsuitable reading order.

Researchers differed, however, on how to organize this process. **FP3** preferred linear operation but emphasized that users “should also be able to skip some steps.” **FP4** preferred “a more freeform” canvas, while **FP5** valued step-by-step control but still wanted to revisit earlier decisions.

Finding 2. Researchers externalized incomplete visual intent through prompts, sketches, references, and generated drafts. These intermediate artifacts let them inspect and progressively revise what should appear, how components relate, and how readers should move through the figure. The evidence supports a guided but revisitable workflow rather than a mandatory sequence.

3.2.3 Evolve: Compare and refine provisional AI-generated candidates. All researchers who used generative AI primarily treated its outputs as provisional candidates that needed to evolve through inspection, selection, reconstruction, and revision. **FP3** described generation as “a little like drawing cards”: after requesting several versions, “one or two” might provide useful inspiration. AI could also critique a manually created figure and suggest possible improvements. **FP4**

valued AI’s creativity and sometimes used generated figures for communication with team members, but stated, “*For figures that go into a paper, I basically redraw them in PowerPoint.*” **FP5** similarly used AI to establish “*the broad direction and overall layout*” before reconstructing the figure manually. These accounts indicate that selecting a promising candidate was only the beginning of the authoring process, not its endpoint.

The transition from an AI-generated candidate to a publication figure was constrained by both visual and scientific problems. **FP1** explained that AI could not reliably understand “*small, detailed modifications*” and expected almost every part of an initial result to require adjustment. **FP3** highlighted unreliable arrows, visual hierarchy, and component relationships. **FP4** found details difficult to control through prompting and ultimately returned to manual drawing. **FP5** reported that AI could copy prompt text into the figure, omit key scientific points, introduce alignment problems, and produce a flattened image that was inconvenient to modify. Even when AI generated a Draw.io file, its content could differ from the displayed preview. Prompt-based regeneration, therefore, offered limited support for evolving a partially correct candidate while preserving the parts that were already valid.

Graphical editability alone was insufficient. **FP3** said that adoption depended on “*whether it is editable*” and whether the result was “*genuinely reasonable.*” **FP2** questioned whether an intermediate representation preserved “*the real data flow,*” while **FP5** warned that a visually plausible icon might not correspond to the meaning of the actual experiment. Participants needed modifications to remain connected to scientific components, relationships, and domain meanings. They also needed to correct a local problem without unintentionally altering validated content elsewhere in the figure.

After viewing the design probe, participants anticipated that direct selection, masking, annotation, and movement could specify edit targets more precisely than chat-based instructions. Such interactions could allow a researcher to identify what should change, preserve non-target content, and inspect the revised result before accepting it.

Finding 3. Researchers evolved promising AI-generated candidates through repeated inspection, reconstruction, and revision. However, existing prompt-based workflows did not support the localized, preservation-aware, and semantically grounded refinement required for publication use.

3.3 Design Objectives

Based on the three formative findings, we derive the following design objectives (DOs) for HiFIGURE:

DO-1 (Envision): The system should allow researchers to retrieve multiple references based on their current authoring intent and determine what each reference, or a selected region within it, should contribute to the methodology figure. Reference roles should remain connected to subsequent authoring decisions.

DO-2 (Externalize): The system should externalize the evolving specification of scientific content as an inspectable intermediate representation, allowing researchers to progressively inspect and revise modules.

DO-3 (Evolve): The system should support bounded revision of provisional candidates, allowing researchers to specify local changes, preserve accepted decisions, and explore, compare, or restore earlier versions.

4 The HiFIGURE System

Guided by the DOs, HiFIGURE supports the progressive intent specification workflow through an evolving provisional representation: a set of inspectable and revisable representations that define what a methodology figure should communicate and how that information should be visually expressed.

Figure 1 illustrates how authors develop this specification through three interconnected activities collaborating with AI support. In **Envision**, authors explore AI-suggested references and select what to borrow from them for layout

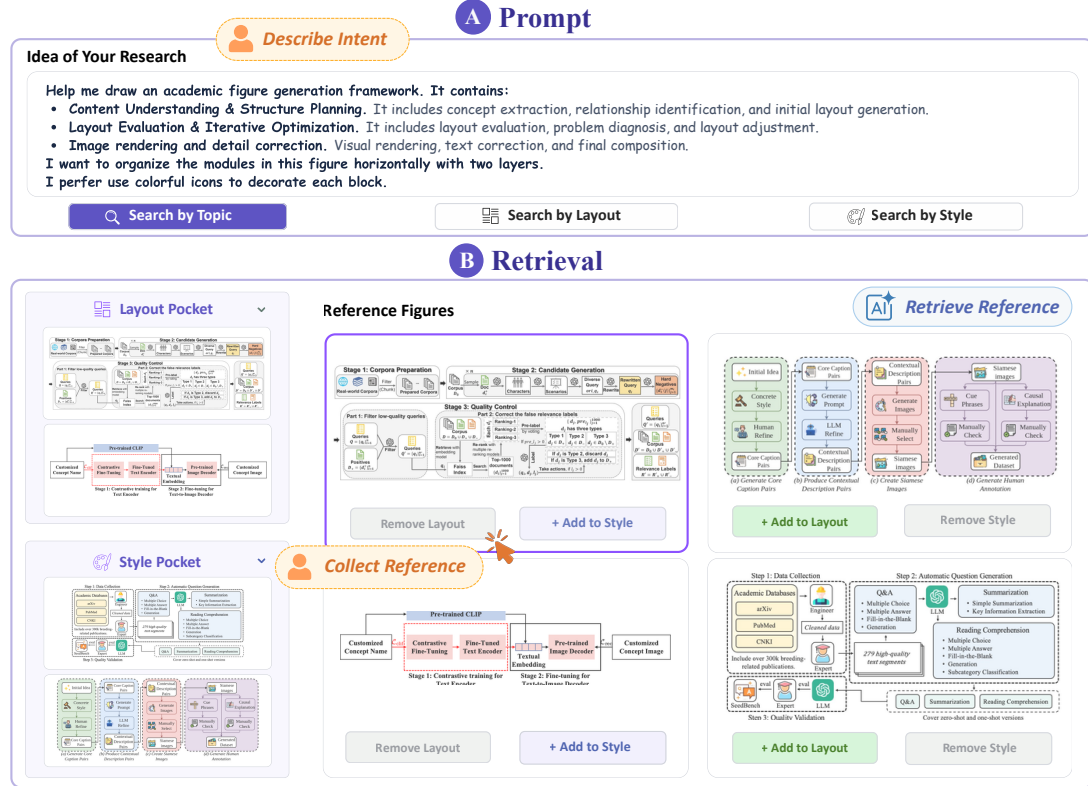


Fig. 2. System implementation of the **Envision** stage in HiFIGURE. (A) *Prompt*: users describe their method and visual preferences and choose a search mode. (B) *Retrieval*: HiFIGURE retrieves reference figures, which users inspect and assign to layout and style roles. Orange and blue callouts indicate human action and AI support, respectively.

and style, establishing an initial visual direction. In **Externalize**, they inspect and revise an AI-generated skeleton and select visual elements extracted from style references, making decisions about scientific structure and visual expression explicit. In **Evolve**, the current specification guides candidate generation, and authors inspect and refine the resulting figures to decide what to retain or revise. Candidates and their associated specifications are preserved in history, allowing authors to compare outcomes and return to earlier configurations. These activities can recur as authors reconsider their choices, progressively developing the representation of their method.

4.1 Envision

To achieve **DO-1**, HiFIGURE supports users in developing a visual direction through two connected modules: *Prompt* (A) and *Retrieval* (B), as shown in Figure 2. Retrieved references provide concrete examples that help researchers develop a visual direction for their own methods.

In **Module A (Prompt)**, researchers describe their method and the visual intent they have already formed. HiFIGURE uses AI to interpret these descriptions and formulate retrieval queries around three aspects of methodology figure authoring. *Search by Topic* retrieves figures with related scientific content, allowing researchers to examine how similar concepts and processes have been represented. *Search by Layout* supports exploration of how components and stages

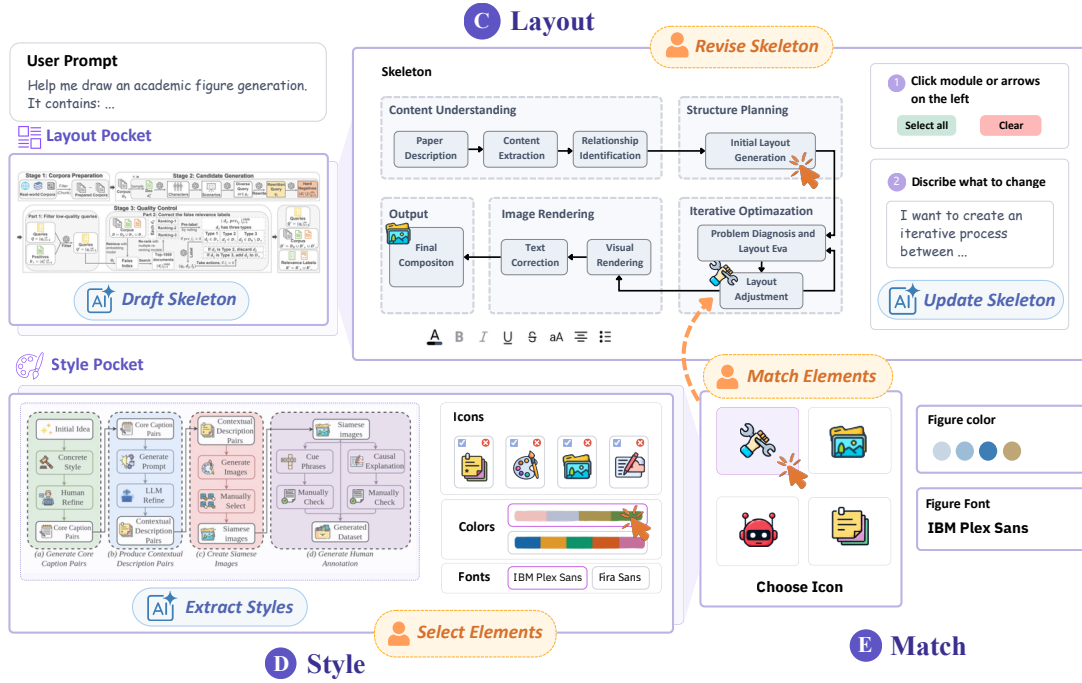


Fig. 3. System implementation of the **Externalize** stage in HiFIGURE. (C) *Layout*: users select a layout reference and revise an AI-generated skeleton. (D) *Style*: users choose a style reference, and HiFIGURE extracts reusable visual elements. (E) *Match*: users assign icons to scientific components and select figure-wide colors and typography. Orange and blue callouts indicate human action and AI support, respectively.

are arranged and how the composition guides reading order. *Search by Style* supports exploration of visual elements such as color, fonts, and icons, which researchers can use to distinguish components and establish emphasis. Researchers select a search mode according to the aspect they are currently working on.

Using the query associated with the selected mode, HiFIGURE retrieves and ranks figures from its reference corpus and presents them in a grid in **Module B (Retrieval)**. AI thus helps turn emerging intent into concrete examples that researchers can inspect and compare. Researchers then decide how these examples could help explain their methods. A useful arrangement may come from one figure, while a suitable visual style may come from another. Therefore, HiFIGURE makes these intended roles explicit through *Add to Layout* and *Add to Style*. A reference can serve either role or both, allowing researchers to specify which aspects they consider useful for presenting their method.

Selected references remain available in the *Layout Pocket* and *Style Pocket*, where researchers can preview or remove them as their direction develops. Layout references guide the organization of the method's components during skeleton generation, while style references guide visual treatment and element extraction. These roles carry the decisions made during **Envision** into **Externalize**. Researchers can begin with partial preferences, use AI-assisted retrieval to explore possible representations, and develop their direction through comparison and selection. They can revisit these choices when a skeleton or a generated candidate helps them assess how the references work with their method.

4.2 Externalize

In *Externalize*, researchers progressively specify how to represent their method. A scientific description may identify the method's components and operations while leaving open how much detail to show, which components to group, and how to organize the explanation visually. To support these decisions (**DO-2**), HiFIGURE exposes the developing figure through an editable skeleton and explicit visual elements (Figure 3). These intermediate representations give researchers something concrete to examine and change while their visual intent is still forming.

The skeleton helps researchers work out the structure of their visual explanation. In **Module C (Layout)**, AI combines the scientific description with a selected layout reference to propose an arrangement of components and connections. Researchers can examine whether an operation needs its own component, whether related components should be grouped, and whether the arrows communicate the intended information flow. By selecting parts of the skeleton and describing changes, researchers can ask AI to revise this proposed representation. The updated skeleton records their more specific choices and provides a basis for further inspection. Structural specification, therefore, develops through examining and revising how the method is represented.

Visual elements require similar decisions about their role in the explanation. A reference may suggest a desirable appearance, but researchers still need to decide which elements fit their method and what those elements should mean. In **Module D (Style)**, AI makes icons, palettes, and font options from references available for inspection and selection. In **Module E (Match)**, researchers assign selected icons to scientific components and choose colors and typography for the figure as a whole. Assigning an icon to a component records its intended meaning within the method. A general preference for a reference's style thus becomes a set of specific choices about how individual components and the overall figure should appear.

These structural and visual decisions remain open as the specification develops. Researchers can reconsider a grouping, revise a connection, or change an icon assignment as they inspect the emerging representation. Externalize makes these decisions explicit enough to guide generation while keeping them available for further revision. The current skeleton and visual settings provide the specification used in **Evolove**, in which a rendered candidate can help researchers assess whether their developing visual explanation communicates the method as intended.

4.3 Evolve

In *Evolve*, researchers use generated figures to refine their visual intent. For methodology figures, this involves examining both how clearly the figure communicates and whether its components and relationships represent the intended method. Inspecting a rendered result may reveal a local problem to correct or a broader structural or stylistic choice to reconsider. To support this process (**DO-3**), HiFIGURE connects candidate generation, targeted revision, and access to earlier specifications through *Candidate* (F) and *History* (G), as shown in Figure 4.

Module F (Candidate) generates the current specification as a complete figure for assessment. AI generates alternatives using the *Prompt*, *Layout* reference, *Skeleton*, *Style* reference, and *Match* settings. Input cards beside the candidate area display these five inputs, allowing researchers to examine the rendered outcomes alongside the choices that guided generation. They can compare how the candidates present the method's components, grouping, and information flow, request another candidate, or select a promising result for further refinement.

Refining a methodology figure requires researchers to communicate both the intended change and its placement. For example, a request to adjust an arrow may concern a specific connection whose direction carries scientific meaning. Researchers can mark regions as edit targets and associate each region with a textual instruction in the **right of Module**

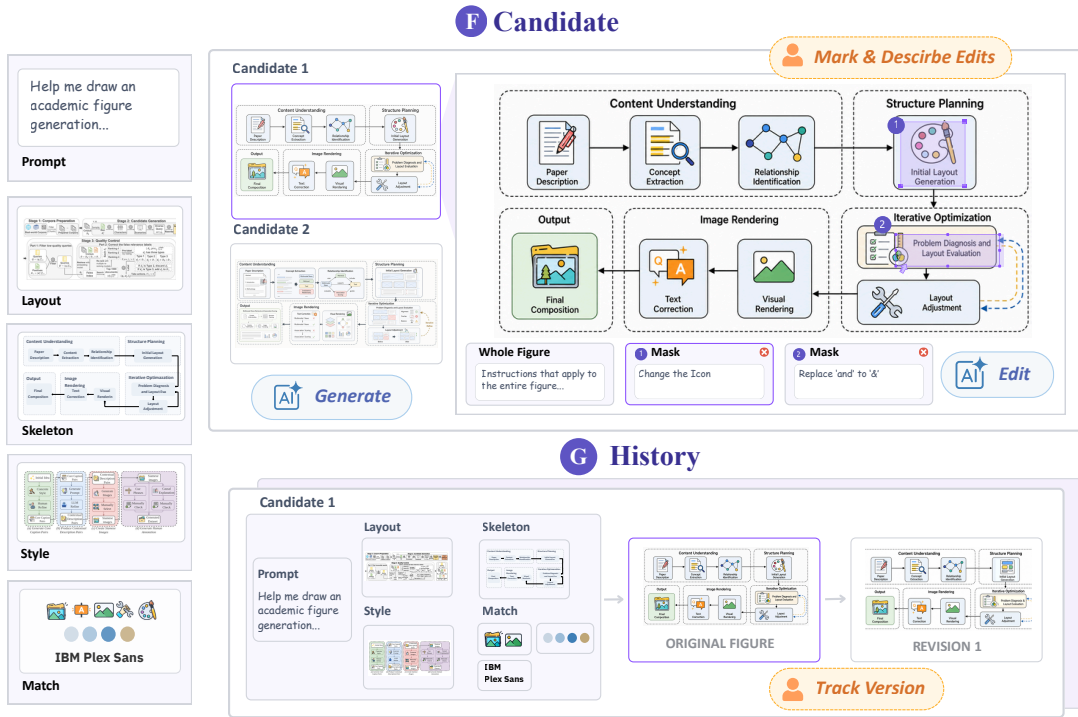


Fig. 4. System implementation of the **Evolve** stage in HiFIGURE. (F) **Candidate**: users compare generated candidates, mark edit targets, and describe local or global changes. HiFigure generates revisions for comparison. (G) **History**: the system associates candidates and revisions with their input specifications, allowing users to revisit earlier states. Orange and blue callouts indicate human action and AI support, respectively.

F (Candidate). Numbered marks connect the selected regions to instructions in the adjacent panel, while a separate *Whole Figure* field accepts changes to the overall composition. These controls let researchers identify the intended scope of a request and describe what should change, including any content or relationships they want to retain.

Inspecting a candidate can also lead researchers to revisit the specification that produced it. **Module G (History)** retains candidates and their revisions together with the associated input specifications, allowing researchers to compare earlier results, restore a previous configuration, and resume authoring from it. They can return to **Envision** to reconsider a reference choice or to **Externalize** to revise component grouping, connections, or visual assignments. The revised specification then guides further generation. In this way, **Evolve** connects decisions made while inspecting rendered figures with the continued development of how the method should be represented.

4.4 Backend Engine

The HiFIGURE backend consists of three interconnected modules that support progressive intent specification, as shown in Figure 5. *Retrieval* supports **envision** by mapping the current intent specification to reference figures. *Layout and Style* supports **externalize** by converting scientific content into an editable structural representation and extracting visual attributes. *Candidate* supports **evolve** by integrating these structured inputs to generate and revise rendered figures.

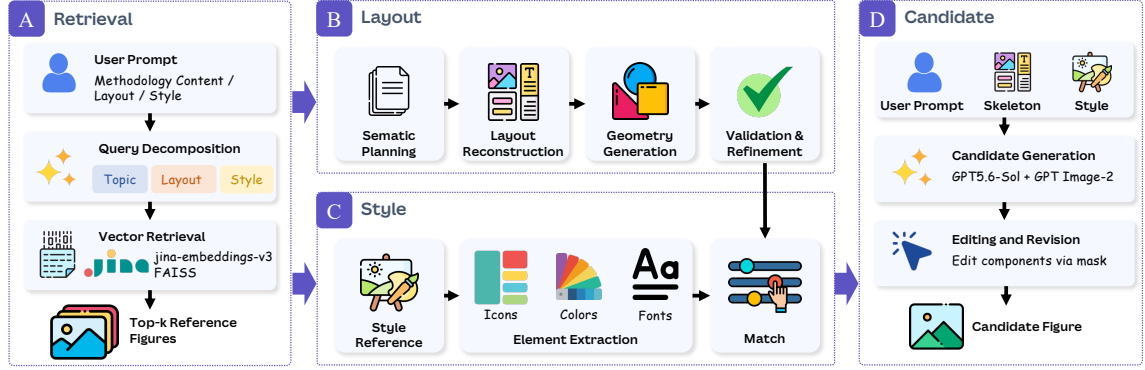


Fig. 5. Overview of the HiFigure backend. *Retrieval* (Panel A) returns reference figures for topic, layout, and style. *Layout* (Panel B) produces an editable figure skeleton. *Style and Match* (Panel C) extracts and assigns visual elements. *Candidate* (Panel D) integrates the resulting content, layout, and style specifications to generate and revise figures.

4.4.1 Envision through Retrieval. We construct a corpus of 4,280 scientific figures, including 3,000 from FigureBench [102] and 1,280 collected from ACL, CVPR, and ICLR 2025. Each figure includes its image, available paper metadata, a content card describing its scientific role and information flow, a design card describing its visual organization, and tags for layout and rendering characteristics. Given the current *Prompt*, GPT-5.4-mini generates structured queries for *Topic*, *Layout*, and *Style*. The *Retrieval* module builds a separate retrieval index for each query type using its corresponding metadata fields. Documents are encoded as normalized 256-dimensional jina-embeddings-v3 vectors [76] and indexed with FAISS [37]. The selected-mode query is embedded and used to rank figures by similarity. Appendix I provides the query-decomposition prompt.

4.4.2 Externalize through Layout and Style. The *Layout* module follows a concept-to-geometry pipeline that transforms scientific content and a layout reference into an editable figure skeleton. An LLM parses the user prompt into components and directed relationships, forming a content graph. The selected layout reference is abstracted into a neutral template that preserves its macro composition. The system maps the content graph onto this template and converts it into concrete component coordinates. Connections are routed around component boxes using orthogonal paths and A* search [27]. A validation pass checks semantic coverage, layout consistency, and connection clarity before exporting the skeleton as editable draw.io XML. Skeleton refinement applies structured updates to the same representation using Claude-Opus-4.8.

The *Style* module converts selected style references into global and component-level visual constraints. The primary style reference defines global properties such as color palette, icon style, and connection appearance. Additional references provide reusable visual elements. Icon extraction uses SAM 2 Hiera-Tiny [67] to segment selected regions into transparent assets and bind them to components in the skeleton. Color extraction quantizes selected regions into compact palettes. Typography matching uses GPT-5.6-sol to analyze visible letterforms and rank up to three visually similar fonts from the installed font library. These results form the *Match* specification, which combines icon bindings with figure-wide color and font constraints. Appendix I provides the corresponding prompts.

4.4.3 Evolve through Candidate. The *Candidate* module converts the current specification into generation conditions for GPT-Image-2. These conditions combine the scientific content, the rasterized skeleton, the layout reference, the

primary style reference, and the additional *Match* constraints. GPT-5.6-sol constructs the generation instructions from these inputs. It first generates a complete figure under the global constraints and then applies the matched visual elements in a second refinement pass. Candidate revision supports component-level editing, masked inpainting, and whole-figure revision over the selected rendered figure. Appendix I provides the corresponding prompts.

5 User Study

HiFIGURE is designed to help researchers articulate, develop, and revise their intent when creating methodology figures. To examine how this support affects the authoring process and its outcomes, we conducted a within-subjects comparative study in which 30 participants used both HiFIGURE and a prompt-based image generation system that serves as the baseline. Our study addressed the following research questions:

- **RQ1:** Does HiFIGURE’s interaction design enhance researchers’ perceived authoring experience?
- **RQ2:** Does HiFIGURE’s Progressive Intent Specification workflow improve the scientific and visual quality of methodology figures?
- **RQ3:** How do researchers iteratively develop intent specifications with HiFIGURE?

5.1 Study Design

Each participant completed two figure creation tasks, using HiFIGURE for one task and the baseline system for the other. To counterbalance system order and task assignment, we used four sequences: B1→H2, H1→B2, B2→H1, and H2→B1, where H and B denote HiFIGURE and the baseline system, respectively, and 1 and 2 denote the two tasks. We distributed participants as evenly as possible across the four sequences.

Participants. We recruited 30 participants across different institutions. The Institutional Review Board approved the study (approval number withheld for anonymous review). All participants confirmed that their research interests are related to artificial intelligence or closely related fields, with particular experience in natural language processing and computer vision. All participants held at least a bachelor’s degree and had experience creating scientific figures for academic publications. They published at least one academic paper (including preprints) within the year before the study and had experience using AI-based image generation tools. All participants signed informed consent forms before the study began.

Baseline System. We compared HiFIGURE with a conventional prompt-based workflow that combined the ChatGPT web interface with a separate reference-image search website. The Baseline used GPT-5.6-sol with the *High* reasoning setting in the ChatGPT web interface and GPT-Image-2 for image generation. Participants also had access to a reference-image search website that used the same scientific figure corpus as HiFIGURE. The website ranked search results using BM25. The baseline also allows for reference retrieval, layout and style adjustments, and iterative modification actions.

Tasks. We designed four scientific-figure creation tasks across two domains: NLP and CV, with two tasks in each domain. Each task described a scientific method through a brief overview, four required modules, an end-to-end workflow, and a set of completion requirements. Participants were instructed to create a methodology figure that represented the method’s key components, their relationships, and the overall information flow. The complete task materials are provided in Appendix B

Procedure. Each study session was conducted in person and lasted approximately 90 minutes. First, participants provided informed consent and completed a background questionnaire. Second, they received an introduction to HiFIGURE and the baseline system, followed by a brief practice session to familiarize themselves with both systems. Third, participants completed the two assigned scientific figure creation tasks and completed a post-task questionnaire after each task. Finally, participants took part in a semi-structured interview to reflect on and compare their experiences with the two systems.

5.2 Measures

We assessed the authoring experience and resulting figures through post-task user self-reported questionnaires, post-task semi-structured interviews, and expert evaluations of figure quality.

User Evaluation. After each task, participants completed a questionnaire assessing both the system and the resulting figure. At the system level, we measured six aspects. *Creativity support* was assessed using six items adapted from the Creativity Support Index (CSI) [14]. *System rating* captured participants' overall evaluation of the system, while *usability* was assessed using UMUX-LITE [45]. *Sense of ownership* and *user agency* were assessed using items adapted from prior work [38]. *Perceived workload* was assessed through ratings of mental effort and task difficulty [61, 72].

At the figure level, we assessed users' perceptions of figure readability and overall figure quality. *Perceived readability* was measured using nine items adapted from three subscales of the Perceived Readability Evaluation for Visualizations (PREVis), including Understandability, Layout Clarity, and Reading Data [13]. *Perceived figure quality* was assessed across five task-specific dimensions adapted from prior literature [52, 59], including usefulness, beauty, faithfulness, completeness, and correctness.

For multi-item measures, we calculated scores as the arithmetic mean of their constituent items. We scored the three PREVis subscales separately by averaging their respective three items. We averaged the five figure-quality items to obtain an overall figure rating. Adapted CSI items used an 11-point response scale from 0 to 10, while mental effort and task difficulty used nine-point scales from 1 to 9. All remaining questionnaire items used seven-point scales from 1 to 7. Appendix C provides the complete questionnaire items and response scales.

Semi-structured Interview. We were particularly interested in how participants perceived the differences between HiFIGURE and the baseline system. The interview, therefore, explored participants' experiences with the two systems in greater depth. We first asked about participants' research backgrounds and their prior practices of using AI for research figure design. We then asked them to compare the two systems in terms of interaction with AI, exploration of alternative designs, perceived control and contribution to the final figure, and overall preference. Finally, participants were invited to describe their ideal AI-assisted figure authoring system and the capabilities they would expect it to provide. The full interview guide is provided in Appendix D.

Expert Evaluation. The quality of the generated figures was also assessed through an expert evaluation. Two experts, each with at least five years of research experience in relevant AI research areas, independently evaluated all 60 figures produced during the study. The evaluation was conducted blind to the system condition, and the figures were presented to each expert in a randomized order. Each figure was assessed using eight dimensions adapted from prior literature [102]: *Aesthetic and Design Quality*, *Visual Expressiveness*, *Professional Polish*, *Clarity*, *Logical Flow*, *Accuracy*, *Completeness*, and *Appropriateness*. These dimensions capture both visual presentation and the communication of

scientific content. Each expert rated every figure using the same five-point rubric. For each expert, an overall figure-quality score was calculated as the arithmetic mean of the eight dimension scores. The dimension definitions and detailed scoring rubric are provided in Appendix E.

5.3 Statistical Analysis

We compared HiFIGURE with the Baseline system using paired observations from 30 participants. User questionnaire responses were paired within participants. Expert ratings were analyzed separately for each expert, pairing the two figures produced by the same participant. For each measure, we examined the normality of within-participant differences using the Shapiro-Wilk test. We applied a two-sided paired t -test when the differences did not significantly deviate from normality ($p > .05$), and a two-sided Wilcoxon signed-rank test otherwise. Holm correction was applied separately across the six system-evaluation measures, nine user-rated figure-evaluation measures, and nine measures for each expert. We report means (M), standard deviations (SD), Holm-adjusted p -values (p_{Holm}), and paired Cohen's d_z . For all comparisons, d_z was calculated as the mean within-participant difference divided by its sample standard deviation. Statistical significance was defined as $p_{\text{Holm}} < .05$.

6 Findings

We organize the findings around the three research questions, examining participants' authoring experiences (RQ1), evaluations of the resulting figures (RQ2), and the development of specifications and perceived control during authoring (RQ3).

6.1 Perceived Authoring Experience (RQ1)

For RQ1, participants reported a more positive authoring experience with HiFIGURE, rating it higher in creativity support, overall system rating, usability, ownership, and agency. We did not detect a significant difference in perceived workload. Figure 6 summarizes these assessments. Composite scores were calculated as the arithmetic mean of their constituent questionnaire items. Further details on the questionnaire items and score calculations are provided in the Appendix F.

Greater creativity support and more positive system evaluations. Participants reported greater creativity support with HiFIGURE than with the Baseline (Figure 6(a)). The six-item mean score was 8.31 ($SD = 1.34$) for HiFIGURE, compared with 5.58 ($SD = 2.02$) for the Baseline ($p_{\text{Holm}} < .001$, $d_z = 1.13$).

Participants also gave HiFIGURE higher System Rating scores ($M = 6.14$, $SD = 0.71$) than the Baseline ($M = 4.20$, $SD = 1.14$; $p_{\text{Holm}} < .001$, $d_z = 1.57$; Figure 6(c)). System Usability scores similarly favored HiFIGURE ($M = 6.00$, $SD = 0.83$) over the Baseline ($M = 4.63$, $SD = 1.50$; $p_{\text{Holm}} < .001$, $d_z = 0.80$; Figure 6(d)).

Stronger ownership and agency. Participants reported a stronger sense of ownership with HiFIGURE ($M = 5.98$, $SD = 0.68$) than with the Baseline ($M = 4.12$, $SD = 1.11$; $p_{\text{Holm}} < .001$, $d_z = 1.51$; Figure 6(e)). They also reported greater user agency with HiFIGURE ($M = 5.82$, $SD = 1.02$) than with the Baseline ($M = 3.87$, $SD = 1.14$; $p_{\text{Holm}} < .001$, $d_z = 1.31$; Figure 6(f)). These ratings address participants' perceived involvement in authoring.

No significant difference in perceived workload. We did not detect a significant difference in perceived workload between HiFIGURE ($M = 4.13$, $SD = 1.96$) and the Baseline ($M = 4.40$, $SD = 1.48$; $p_{\text{Holm}} = .588$, $d_z = -0.10$; Figure 6(b)).

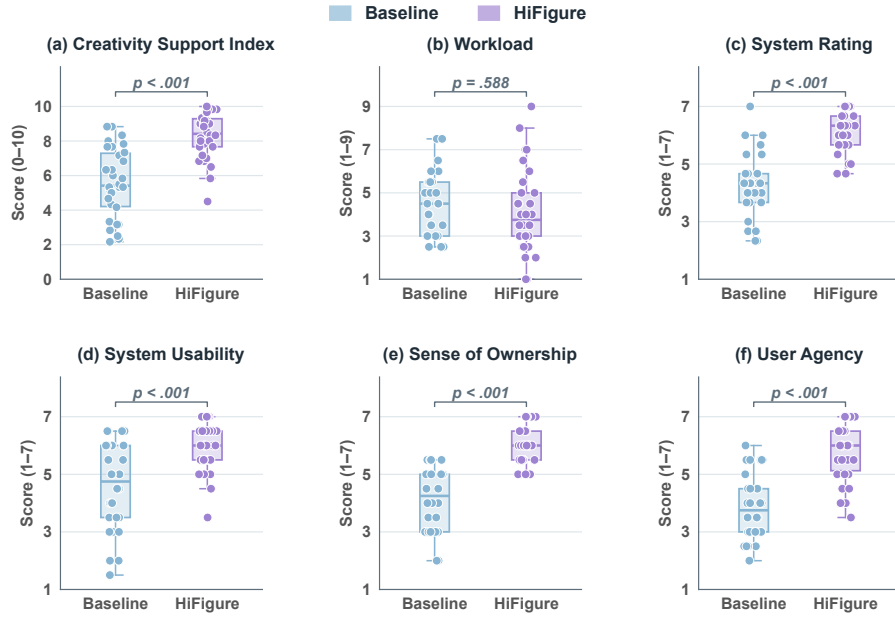


Fig. 6. User assessments of the Baseline and HiFIGURE ($n = 30$ pairs). Panels show (a) creativity support (six-item mean), (b) perceived workload, (c) System Rating, (d) System Usability, (e) Sense of Ownership, and (f) User Agency.

The positive answer to RQ1, therefore, concerns perceived creativity support, system evaluations, ownership, and agency; the workload result does not establish a reduction in perceived effort.

6.2 Scientific and Visual Quality of Figures (RQ2)

For RQ2, participant and expert evaluations consistently favored figures created with HiFIGURE across the assessed scientific and visual quality dimensions. Participants rated their figures as more readable and higher in overall quality, and both experts assigned higher overall and individual dimension scores. These findings support a positive answer to RQ2 for figures produced using the complete HiFIGURE system. Further details are provided in the Appendix G.

6.2.1 Participants rated figures as more readable and higher in quality. Figure 7 presents participant evaluations of readability, overall figure quality, and five individual quality dimensions.

Higher user perceived readability. We analyzed the three retained PREVis subscales separately, calculating each subscale score as the mean of its three items (Figure 7(a)). Participants rated HiFIGURE figures higher on *Understandability* ($M = 5.96$, $SD = 0.73$) than Baseline figures ($M = 4.43$, $SD = 1.11$; $p_{\text{Holm}} < .001$, $d_z = 1.33$). *Layout Clarity* also favored HiFIGURE ($M = 5.81$, $SD = 0.96$) over the Baseline ($M = 3.87$, $SD = 1.30$; $p_{\text{Holm}} < .001$, $d_z = 1.32$). Similarly, HiFIGURE received higher *Reading Data* scores ($M = 6.04$, $SD = 0.73$) than the Baseline ($M = 4.18$, $SD = 1.09$; $p_{\text{Holm}} < .001$, $d_z = 1.38$).

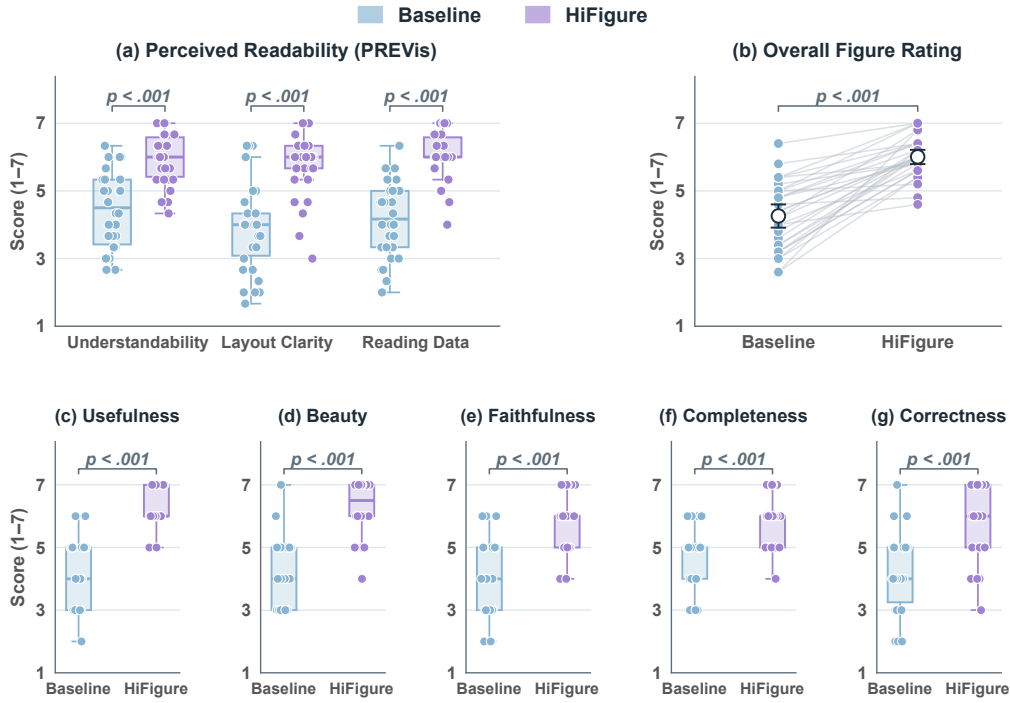


Fig. 7. Participant evaluations of figures produced with the Baseline and HiFigure ($n = 30$ pairs). Panels show (a) PREVis, (b) Overall Figure Rating (the mean of five quality items), (c) Usefulness, (d) Beauty, (e) Faithfulness, (f) Completeness, and (g) Correctness. Panel (a) shows score distributions and individual observations; lines in (b) connect scores from the same participant.

Higher user perceived overall quality. Figures produced by HiFIGURE received higher overall ratings ($M = 6.01$, $SD = 0.60$) than Baseline figures ($M = 4.26$, $SD = 1.00$; $p_{\text{Holm}} < .001$, $d_z = 1.78$; Figure 7(b)). All 30 participants gave their HiFIGURE figure a higher overall score than their corresponding Baseline figure.

The overall difference was accompanied by higher ratings on all five quality dimensions (Figure 7(c–g)). HiFIGURE figures were rated as more useful ($M = 6.20$, $SD = 0.66$ vs. $M = 4.13$, $SD = 1.07$; $p_{\text{Holm}} < .001$, $d_z = 2.04$) and more beautiful ($M = 6.30$, $SD = 0.84$ vs. $M = 4.23$, $SD = 1.07$; $p_{\text{Holm}} < .001$, $d_z = 1.76$). Faithfulness ratings were also higher ($M = 5.87$, $SD = 0.86$ vs. $M = 4.03$, $SD = 1.40$; $p_{\text{Holm}} < .001$, $d_z = 1.20$), as were Completeness ($M = 5.87$, $SD = 0.78$ vs. $M = 4.80$, $SD = 0.92$; $p_{\text{Holm}} < .001$, $d_z = 0.93$) and Correctness ($M = 5.80$, $SD = 1.10$ vs. $M = 4.10$, $SD = 1.30$; $p_{\text{Holm}} < .001$, $d_z = 1.03$).

6.2.2 Experts rated figures higher across scientific and visual dimensions. The expert evaluations provided further evidence for RQ2. Two experts evaluated the 60 figures produced by the 30 participants across eight dimensions: Aesthetic and Design Quality, Visual Expressiveness, Professional Polish, Clarity, Logical Flow, Accuracy, Completeness, and Appropriateness. We analyzed each expert's ratings separately, pairing the two figures produced by the same participant. Each expert's overall figure-quality score was the arithmetic mean of the eight dimension scores.

Both experts rated HiFIGURE figures higher than Baseline figures in overall quality and across all eight individual dimensions (Figure 8).

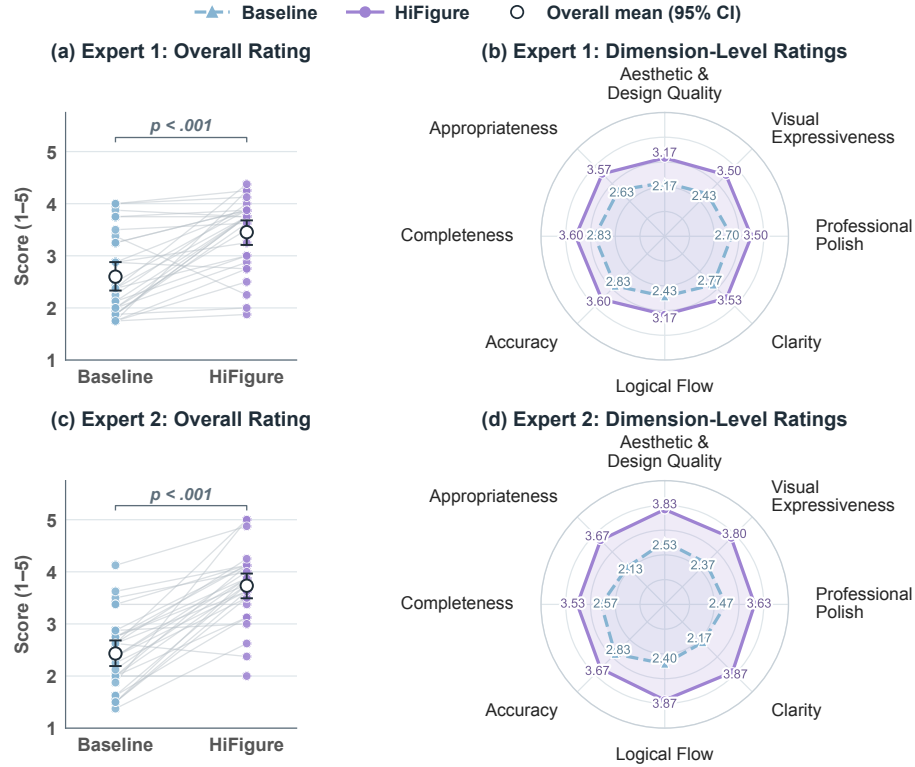


Fig. 8. Expert evaluations of figure quality. (a,b) Expert 1. (c,d) Expert 2. Panels (a,c) show overall scores, calculated as the mean of eight dimensions; lines connect the two figures produced by the same participant.

Higher overall figure quality. For Expert 1, the mean score was 3.45 ($SD = 0.67$) for HiFIGURE and 2.60 ($SD = 0.78$) for the Baseline ($p_{\text{Holm}} < .001$, $d_z = 1.03$; Figure 8(a)). Expert 2 rated HiFIGURE figures at 3.73 ($SD = 0.68$), compared with 2.43 ($SD = 0.70$) for the Baseline ($p_{\text{Holm}} < .001$, $d_z = 1.64$; Figure 8(c)).

Higher ratings across individual dimensions. Both experts rated HiFIGURE figures significantly higher across all eight dimensions (all $p_{\text{Holm}} < .01$; Figure 8(b,d)). Detailed statistics are reported in Appendix H. Participant and expert evaluations thus converged on HiFIGURE across dimensions addressing scientific content and visual presentation.

6.3 Progressive Intent Specification and Perceived Control (RQ3)

RQ3 examines how researchers developed and revised intent specifications with HiFIGURE, and what shaped their perceived control. We combine interaction logs, within-participant correlations, and interviews to characterize how participants revisited decisions, shifted between revision activities, and specified what the AI should preserve.

6.3.1 Intent Specification Was Non-linear. Figure 9 shows that 28 of the 30 participants moved from a later phase to an earlier phase and then returned to the later phase. The median sequence contained nine visits (range: 3–36). At least 25 participants completed such a return before generating their first candidate. Revisiting, therefore, occurred while participants were developing an initial specification, not only when correcting a generated figure. As P05 described,

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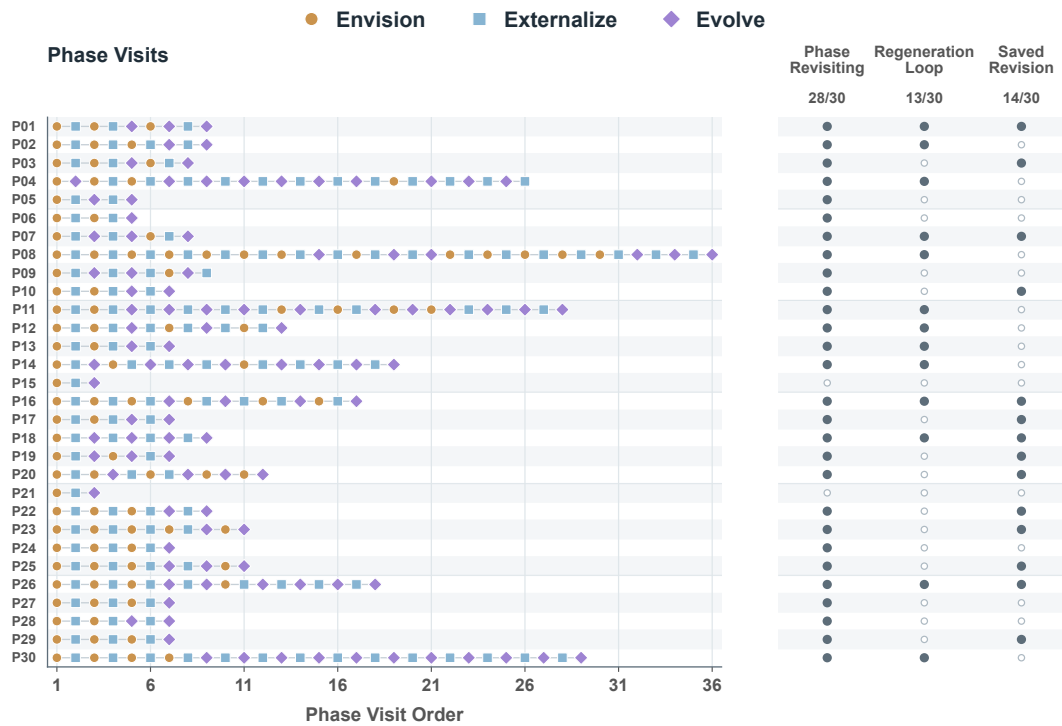


Fig. 9. Authoring-phase sequences in HiFIGURE. Consecutive records in the same phase are merged into one visit; the right-hand columns summarize revisits and revisions.

the ability to “go back to the previous step at any time” or branch from an earlier version helped keep prior decisions available for reconsideration.

6.3.2 Participants Revised Specifications and Rendered Figures. Figure 10(a) shows broader activity coverage in HiFIGURE. Edit activity appeared for 23 participants in HiFIGURE and 18 in the Baseline; Retrieval for 30 and 22; Layout for 18 and 11; and Style for 29 and 15, respectively. The largest difference concerned Style. Participant-level counts nevertheless varied: for example, **P16** and **P28** each made 14 requests in HiFIGURE, whereas **P19** made more requests in the Baseline.

Participants also revised at different representational levels. Thirteen modified structure, style, or icons between consecutive generations, 14 saved revisions to rendered images, and five did both. For example, after generating a candidate, **P13** returned to an earlier skeleton and separated the Active View Controller from the Evidence Integrator before generating the next candidate. Revision thus involved both visual refinement and decisions about how scientific components should be represented.

6.3.3 How Visual Support Relates to Figure Quality. To examine how visual support use relates to figure quality, we computed participant-level condition differences ($\Delta = \text{HiFIGURE} - \text{Baseline}$) in operation coverage and expert ratings, then correlated these differences using Spearman’s ρ .

The associations were strongest for activities supported by explicit visual references and reusable style elements in Figure 11. Greater increases in Style activity were associated with larger gains in polish ($\rho = .40$), expressiveness

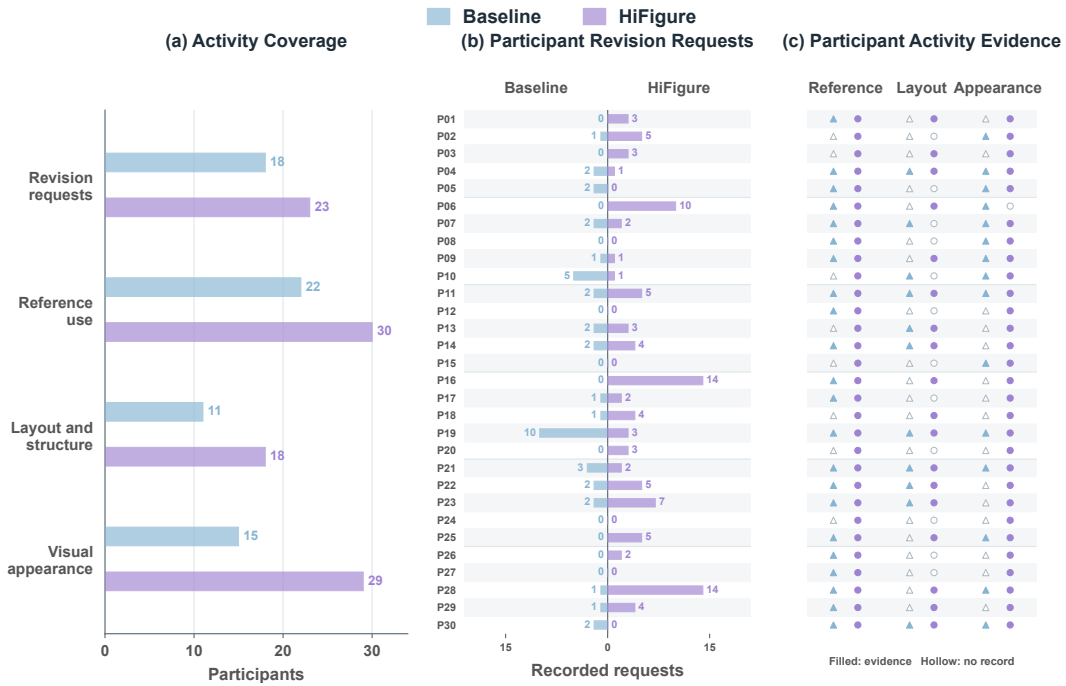


Fig. 10. Authoring operations in HiFigure and the Baseline: (a) participant coverage by category and (b) participant-level revision-request counts.

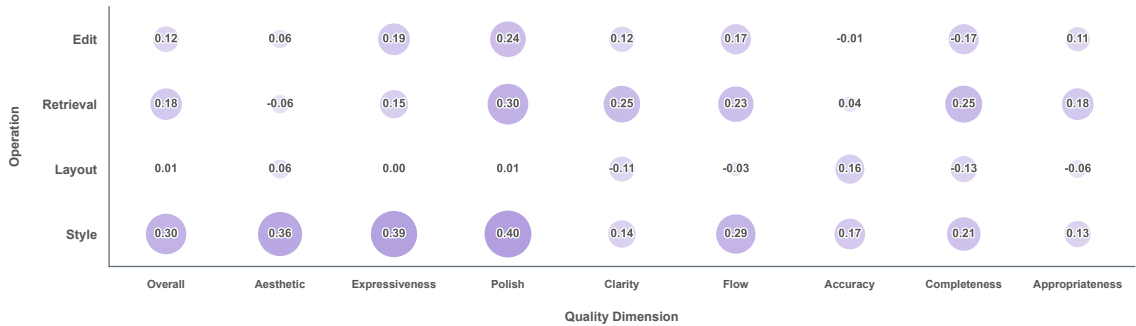


Fig. 11. Spearman correlations across participants between within-participant condition differences ($\Delta = \text{HiFigure} - \text{Baseline}$) in operation coverage and expert-rated figure quality.

($\rho = .39$), and aesthetics ($\rho = .36$). One possible explanation is that making icons, colors, and typography available for inspection and reuse helps researchers coordinate visual choices and communicate concepts more effectively. Retrieval showed more modest associations with polish ($\rho = .30$), clarity ($\rho = .25$), and completeness ($\rho = .25$), consistent with the potential value of concrete references for guiding composition and content presentation. Edit and Layout showed weaker, mixed associations. These patterns suggest that HiFigure's visual supports may help researchers translate evolving intent into visual choices associated with higher figure quality.

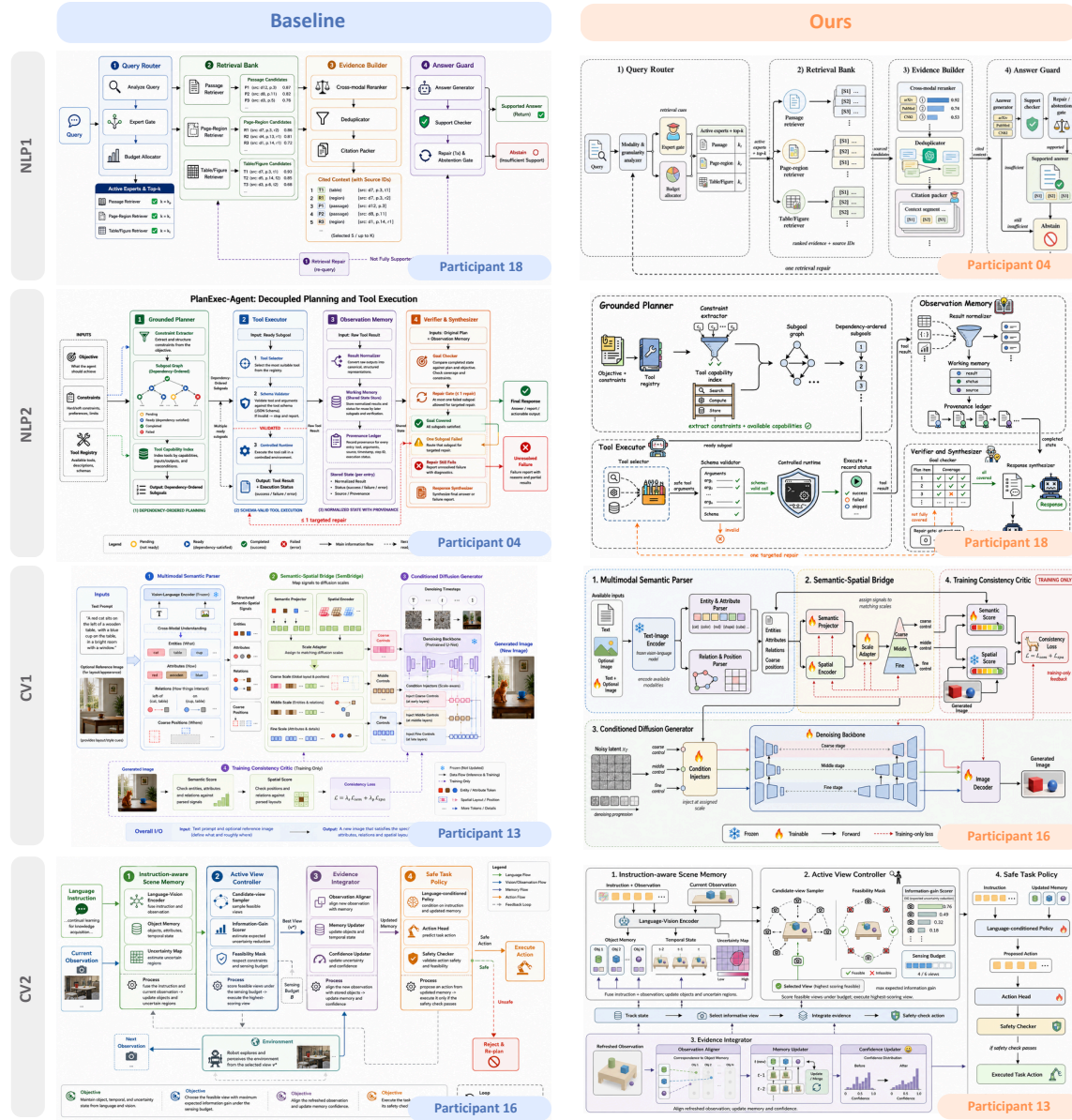


Fig. 12. Methodology figures generated with the Baseline (left) and HiFIGURE (right). Rows correspond to NLP1, NLP2, CV1, and CV2. Each row shows the same task completed by different participants, identified below each figure.

6.3.4 Control Depended on Delegation and Preservation. Interviews help explain these mixed patterns. Participants valued direct editing when the intended change was already clear. **P08** said, “I don’t need to tell it; I can do it myself.” In contrast, participants used AI assistance when they could identify a problem without knowing how to redraw it. Control also depended on limiting the scope of an AI revision. **P10** requested less text but then found that a scientific

component was missing; **P03** corrected an arrow’s origin and direction; and **P23** asked to remove unnecessary dots while preserving the arrows. Together, the logs, ratings, and interviews show that perceived control came from being able to revisit a decision, choose how to intervene, and state what should remain unchanged.

6.4 Examples of generated figures

Figure 12 presents example outputs across the four study tasks, illustrating differences in layout, component grouping, and visual detail. Each row shows the same task represented by different participants using the Baseline and HiFIGURE.

7 Discussion

7.1 When Progress Means Going Back? Rethinking Intent Specification

HiFIGURE is organized around **Envision**, **Externalize**, and **Evolve**, but participants rarely moved through these activities in order. 28 of the 30 participants returned to an earlier phase, and at least 25 did so before generating their first candidate figure (Section 6.3.1). **P05** valued being able to “*go back to the previous step at any time, look at it again, or branch off a new version from a particular version node.*” The logs show that these returns were not simply attempts to repair a poor generation. They also occurred while participants were still working out what the figure should contain and how to arrange its elements. A figure-creation system should preserve this intermediate work so that an earlier decision can be revisited without forcing the researcher to start over.

IntentTagger and Spellburst also let users move between representations or branch into alternatives instead of following a fixed sequence [2, 24]. Methodology figures add a particular complication: choices about scientific content and visual presentation are often entangled. A researcher might choose a reference for its layout, build a skeleton around it, and only later discover that another arrangement explains the method better. We observed this kind of reconsideration with the Baseline as well, so it appears to be part of figure creation rather than an artifact of HiFIGURE’s workflow. In HiFIGURE, persistent references, editable skeletons, and linked versions give such intermediate choices a durable form. This extends earlier ideas of reification and reuse in visual interfaces [6], as well as the inspection and redirection of intermediate AI outputs in AI Chains [93]. Here, however, the intermediate output also records how the researcher’s account of the method is developing.

Returning to an earlier version is only part of the problem. A researcher may want to revisit the layout while keeping a component grouping intact. Work on parallel alternatives, linked editing, and layout exploration under high-level constraints offers useful starting points for carrying selected decisions across versions [28, 82]. A history view could also expose differences between intent specifications, much as prior work has supported visual comparison of slide-deck versions [20]. Selective carryover and explicit comparison would make it easier to recover an earlier choice without losing useful work that came afterward.

We use *Progressive Intent Specification* to describe this accumulation and revision of explicit decisions. “Progressive” does not imply a one-way march toward an increasingly fixed specification. In our study, progress often took the form of reopening an earlier choice after a new representation changed how the researcher understood the figure. This pattern echoes accounts of problem–solution co-evolution, in which designers repeatedly revise their understanding of both the problem and its possible solutions [18].

7.2 What Should the Figure Say? Shaping the Method’s Visual Story

At the outset, researchers may know what they need to communicate without knowing what the figure should look like. Turning that broad goal into visual intent is difficult [77], particularly when the interaction starts with a text prompt. **P06** described the difficulty directly: “*First, I need to know what style I want and what words describe it. If I don’t know that, it’s hard for it to generate it.*” Visual references offered a different way to begin. **P18**, for instance, valued encountering figures outside their own field. These examples exposed possible ways to structure an explanation, as well as styles to draw on. GenQuery and CreativeConnect similarly use visual search and reference recombination to support idea development [15, 75]. This may partly account for HiFIGURE’s higher creativity-support ratings ($M = 8.31$ vs. 5.58 for the Baseline, $p_{\text{Holm}} < .001$; Section 6.1). Participants could inspect possibilities before they had the vocabulary to request them, consistent with work on how external representations reveal previously unnoticed possibilities [40, 81].

Visual references and reusable style elements may also support the visual presentation of scientific content. In HiFIGURE, explicit choices about icons, colors, and typography give authors concrete ways to coordinate appearance and distinguish component roles. This offers one possible explanation for the exploratory associations between greater Style activity relative to the Baseline and gains in polish, expressiveness, and aesthetics (Figure 11). Such support may help authors develop a coherent visual language as their intent evolves, although these correlations do not establish the effects of individual system features.

Candidate figures also helped participants inspect and revise how scientific components were represented. After inspecting one candidate, **P13** returned to an earlier skeleton and separated the Active View Controller from the Evidence Integrator. The next candidate was generated from a skeleton in which the Evidence Integrator appeared as a distinct component. In this case, the rendered figure exposed the consequences of an earlier structural choice, and the skeleton provided a place to revise it. References, skeletons, and candidates thus supported different moments in the same process: references suggested directions, skeletons articulated a developing explanation, and candidates tested that explanation.

Several participants wanted the AI to take a more active role in this process. **P22** said, “*I think an AI needs to assess what you enter and offer criticism and corrections.*” **P21** asked for explanations of the modules and their logic, while **P24** wanted follow-up questions about the initial request. These comments point toward concrete forms of assistance. The system might ask why two operations share a block, show another way to depict a dependency, or flag information that could disappear through simplification. Such support draws on embedded design critics [22] and mixed-initiative interaction [35]. Critique should support researchers in examining emerging decisions while preserving their ability to inspect, reject, or revise a suggestion [1, 22]. Questions and critiques could complement HiFIGURE’s existing representations by helping researchers test whether an emerging specification clearly explains the method.

7.3 Who Dominates Changes? Negotiating Control in Human-AI Figure Revision

Participants did not consistently favor either manual editing or AI-assisted revision. Their choice depended on the change at hand. When the desired change was already clear, describing it to an AI could be slower than making it directly. **P08** put this plainly: “*I don’t need to tell it; I can do it myself.*” **P14** identified a related difficulty: “*There may be just one small part that needs changing, but it does not know exactly where I want to make the change.*” The interaction logs reflect this mixed practice. Thirteen participants modified a specification between consecutive generations, 14 saved image revisions, and five did both (Section 6.3). Direct manipulation helped move or regroup known components; AI assistance was more attractive when a participant could locate a problem but was unsure how to redraw it. DirectGPT,

PromptCharm, and Keyframer similarly combine language-based generation with graphical interaction and editable outputs [58, 84, 88]. HiFIGURE’s editable skeletons and region-based image revision support the same shift in control as the task changes.

The revision examples also show that specifying what should change is not enough. Researchers assumed that accepted parts of the figure would remain intact. **P03** corrected the origin and direction of an arrow; **P23** wanted unnecessary dots removed but the arrows preserved. With the Baseline, **P10** requested less text while retaining complete content, yet the revision dropped a component. Each request, therefore, carried an implicit preservation constraint. Selecting a region tells the AI where to act, but may not reveal which components, relations, or details in that region must survive the edit. Semantic Commit addresses related dependencies by considering how a change to one part of an intent specification affects information elsewhere [86]. PaperBanana-Interact instead checks a proposed revision against earlier requests and source context [94].

These cases suggest a practical protocol for AI-assisted revision. When an edit may affect accepted content, the system should first show the proposed scope and any likely consequences outside it. The researcher can then protect a component or relationship, clarify the request, or deliberately reopen an earlier decision. This brief exchange would preserve the convenience of delegation while making its limits explicit. The central issue is not how much autonomy to give the AI in general, but whether the researcher can see and adjust what is being delegated in a particular edit.

7.4 Limitations and Future Work

The interviews capture participants’ initial experience with HiFIGURE. Three participants preferred the Baseline’s conversational interaction, with **P04** and **P10** explicitly attributing this preference to familiarity. **P28** instead valued its direct instruction and response process. All three nevertheless preferred HiFIGURE for exploring visual alternatives. These accounts suggest that workflow preferences depend on both prior experience and the activity being supported. Longer-term use would help establish whether the reported benefits persist as researchers become familiar with HiFIGURE, which controls they continue to use, and how the desired balance between guidance and independent action changes over time.

We evaluated HiFIGURE as a complete system, so the comparison does not isolate the contributions of reference retrieval, editable skeleton, style selection, and local revision. Our logs document phase returns and specification changes, but these records do not directly measure how a researcher’s intent developed or explain every return. Feature-focused studies and retrospective accounts linked to specific interaction episodes could help clarify when each form of support contributes to a decision. Distinguishing effort spent exploring, specifying, checking, and repairing would also help interpret the agency gains alongside the absence of a detected workload difference.

The figure evaluation combined researchers’ self-assessments with ratings from two experts. These measures provide evidence of perceived and judged quality, while leaving open whether readers understand the resulting figures more accurately. Reader studies could assess comprehension of components, relationships, and information flow, and a broader set of experts could capture variation in disciplinary expectations. Future work should examine whether generation and subsequent edits preserve the components and relationships specified in the skeleton, particularly when structural and visual changes interact.

Interview feedback suggests opportunities to adapt assistance to researchers’ needs. Future versions could let users skip familiar steps, request explanations of unfamiliar modules, and ask for critical feedback when needed. More discoverable editing controls and editable final exports could support precise revisions while preserving opportunities to delegate broader changes to AI.

8 Conclusion

We presented HiFIGURE, a human-AI collaborative system for progressively specifying methodology figures. A formative study informed its support for three interdependent activities: *Envision*, *Externalize*, and *Evolve*. In a within-subject study with 30 participants, HiFIGURE received higher ratings for creativity support, usability, ownership, and agency than a prompt-based AI workflow. Its figures also received higher participant and expert ratings. Logs and interviews showed how participants revisited visual decisions and differed in their preferred forms of control. HiFIGURE takes a step toward closing the gap between knowing a method and making its logic visible, helping researchers move from “I know my method, but not how to show it” to “This is my method, and this is how I choose to tell its story.”

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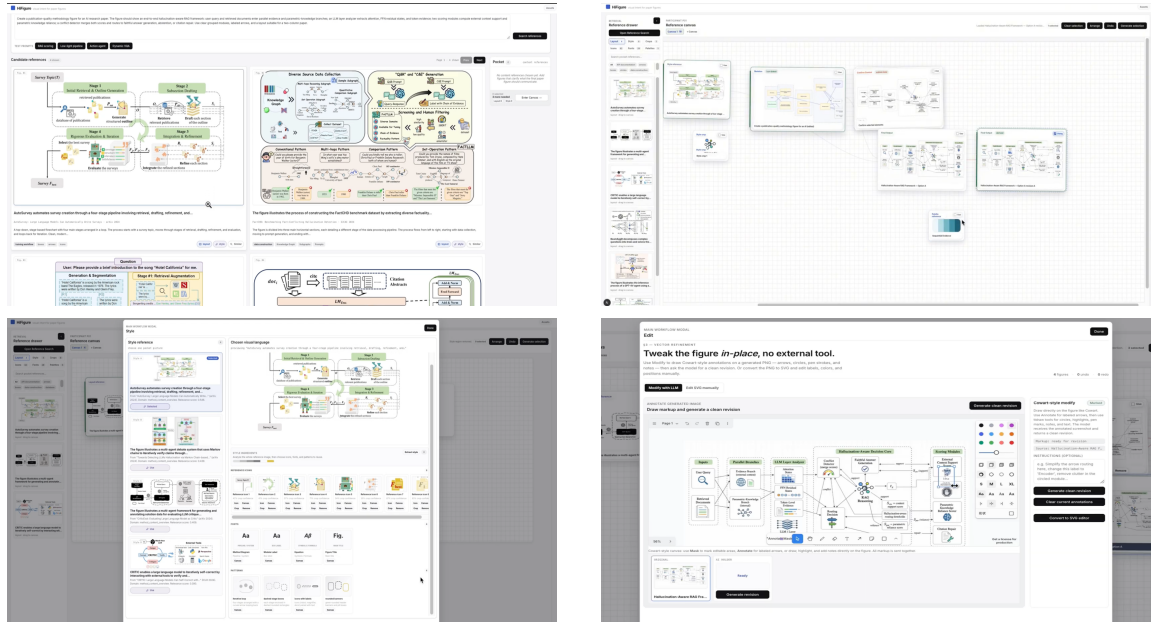


Fig. 13. Early HiFigure prototype used in the formative study. From top left to bottom right: reference search and selection; a canvas for organizing references, intermediate representations, and figure candidates; extraction and selection of reusable style elements; and figure revision through annotations and textual instructions.

A Formative Study System Version

Figure 13 shows the early HiFigure prototype used as a design probe in the formative study. To reduce priming, we demonstrated it after discussing participants' existing figure-authoring practices. Participants then compared the prototype with their workflows and commented on its strengths, limitations, and potential improvements.

B Study Task Materials

This appendix presents the four scientific figure creation tasks used in the study. Participants selected either the natural language processing (NLP) or computer vision (CV) domain based on their familiarity and completed both tasks in the selected domain. The system assigned to each task varied according to the counterbalancing sequence.

Participants selected either the natural language processing (NLP) or computer vision (CV) domain and completed both tasks in the selected domain. The following task descriptions were presented to participants.

B.1 NLP Task 1: RouteFuse-RAG

RouteFuse-RAG: Adaptive Cross-modal Evidence Routing

Method Overview. A query-aware RAG system selects text, page-region, and table/figure retrievers, merges their evidence into a cited context, and returns an answer only when its claims are supported.

Required Modules.

(1) Query Router

Purpose. Choose evidence sources and a top- k budget from the query.

Components. Modality and granularity analyzer, expert gate, and budget allocator.

Process. Extract retrieval cues $\rightarrow$ output active experts and top- k values.

(2) Retrieval Bank

Purpose. Search the selected sources and preserve provenance.

Components. Passage retriever, page-region retriever, and table/figure retriever.

Process. Search each active index $\rightarrow$ return ranked evidence with source IDs.

(3) Evidence Builder

Purpose. Produce one compact, non-redundant cited context.

Components. Cross-modal reranker, deduplicator, and citation packer.

Process. Rerank and merge candidates $\rightarrow$ pack evidence with citation IDs.

(4) Answer Guard

Purpose. Generate the answer and enforce claim support.

Components. Answer generator, support checker, and repair and abstention gate.

Process. Draft and check cited claims $\rightarrow$ return the answer, repair retrieval once, or abstain.

End-to-end Method. The Query Router outputs active retrievers and top- k values. The Retrieval Bank returns sourced candidates, and the Evidence Builder turns them into a cited context. The Answer Guard returns a supported answer. Otherwise, it requests one retrieval repair and then abstains if support remains insufficient.

Completion Requirements.

- (1) Output active retrieval experts and their top- k values.
- (2) Retrieve passage, page-region, and table/figure evidence with source IDs.
- (3) Rerank and deduplicate evidence before building the cited context.
- (4) Return only supported claims, with one retrieval repair allowed before abstention.

Participant Instruction. Please use the assigned system to generate a scientific figure that satisfies the requirements above. Take a screenshot of the selected figure or copy it into the designated area.

B.2 NLP Task 2: PlanExec-Agent

PlanExec-Agent: Decoupled Planning and Tool Execution

Method Overview. A modular LLM agent turns an objective into an ordered plan, binds ready steps to valid tool calls, records tool results in shared state, and performs one bounded repair before answering.

Required Modules.

(1) Grounded Planner

Purpose. Create an ordered plan from the objective, constraints, and tool registry.

Components. Constraint extractor, subgoal graph, and tool capability index.

Process. Extract constraints and available capabilities $\rightarrow$ output dependency-ordered subgoals.

(2) Tool Executor

Purpose. Bind each ready subgoal to a safe, schema-valid call.

Components. Tool selector, schema validator, and controlled runtime.

Process. Select and validate a tool call → execute it and record its status.

(3) Observation Memory

Purpose. Convert tool outputs into reusable state with provenance.

Components. Result normalizer, working memory, and provenance ledger.

Process. Normalize the tool result → update the state with its source and status.

(4) Verifier and Synthesizer

Purpose. Check goal coverage and produce the final response.

Components. Goal checker, repair gate, and response synthesizer.

Process. Compare the completed state with the plan → repair one failed subgoal or synthesize the response.

End-to-end Method. The Grounded Planner outputs dependency-ordered subgoals. For each ready subgoal, the Tool Executor records a validated result in Observation Memory. The Verifier and Synthesizer repairs at most one failed subgoal and then returns the response or reports the unresolved failure.

Completion Requirements.

- (1) Output dependency-ordered subgoals from the objective and constraints.
- (2) Validate each tool and its arguments before execution.
- (3) Record normalized results, statuses, and provenance in shared state.
- (4) Check goal coverage and allow at most one targeted repair before responding.

Participant Instruction. Please use the assigned system to generate a scientific figure that satisfies the requirements above. Take a screenshot of the selected figure or copy it into the designated area.

B.3 CV Task 1: SemBridge

SemBridge: Semantic-Spatial Conditioning for Diffusion

Method Overview. A frozen vision-language model extracts entities, attributes, relations, and coarse positions. A bridge injects these signals into matching diffusion scales, while a training-only critic scores semantic and spatial agreement.

Required Modules.

(1) Multimodal Semantic Parser

Purpose. Convert text and an optional image into structured generation signals.

Components. Text-image encoder, entity and attribute parser, and relation and position parser.

Process. Encode the available modalities → output entities, attributes, relations, and coarse positions.

(2) Semantic-Spatial Bridge

Purpose. Map parsed signals to diffusion-compatible controls.

Components. Semantic projector, spatial encoder, and scale adapter.

Process. Project semantic and spatial signals → assign them to coarse, middle, or fine scales.

(3) Conditioned Diffusion Generator

Purpose. Generate an image using scale-matched controls.

Components. Denoising backbone, condition injectors, and image decoder.

Process. Inject controls at their assigned scales → denoise and decode the image.

(4) Training Consistency Critic

Purpose. Supervise semantic and spatial fidelity during training only.

Components. Semantic score, spatial score, and consistency loss.

Process. Compare the generated image with the parsed signals → return the combined training loss.

End-to-end Method. The Multimodal Semantic Parser outputs entities, attributes, relations, and coarse positions. The Semantic-Spatial Bridge assigns these signals to matching diffusion scales, and the generator produces the image. During training only, the Training Consistency Critic returns semantic and spatial losses to update the bridge and generator.

Completion Requirements.

- (1) Output entities, attributes, relations, and coarse positions from the available inputs.
- (2) Map parsed signals to coarse, middle, or fine diffusion controls.
- (3) Inject every control at its assigned denoising scale.
- (4) Use semantic and spatial consistency loss during training only.

Participant Instruction. Please use the assigned system to generate a scientific figure that satisfies the requirements above. Take a screenshot of the selected figure or copy it into the designated area.

B.4 CV Task 2: ActiveView-VLA

ActiveView-VLA: Uncertainty-guided Embodied Perception

Method Overview. A vision-language-action agent tracks objects and uncertainty, selects one feasible viewpoint with the highest expected information gain, updates memory from the new observation, and executes a safety-checked task action.

Required Modules.

(1) Instruction-aware Scene Memory

Purpose. Maintain object, temporal, and uncertainty state from language and vision.

Components. Language-vision encoder, object memory, and uncertainty map.

Process. Fuse the instruction and current observation → update objects and uncertain regions.

(2) Active View Controller

Purpose. Acquire the feasible view with the largest expected uncertainty reduction.

Components. Candidate-view sampler, information-gain scorer, and feasibility mask.

Process. Score feasible views under the sensing budget → execute the highest-scoring view.

(3) Evidence Integrator

Purpose. Merge the refreshed observation into persistent state.

Components. Observation aligner, memory updater, and confidence updater.

Process. Align the new observation with stored objects → update memory and confidence.

(4) Safe Task Policy

Purpose. Choose and validate the embodied task action.

Components. Language-conditioned policy, action head, and safety checker.

Process. Propose an action from updated memory → execute it only if the safety check passes.

Table 1. Items included in the post-task questionnaires for GPT Baseline and HiFigure System.

ID	Construct	Item	Scale
CSI1	Creativity Support Index	It was easy for me to explore many different ideas, options, designs, or outcomes using [System].	0–10
CSI2	Creativity Support Index	[System] was helpful in allowing me to track different ideas, outcomes, or possibilities.	0–10
CSI3	Creativity Support Index	I was able to be very creative while doing the activity inside [System].	0–10
CSI4	Creativity Support Index	[System] allowed me to be very expressive.	0–10
CSI5	Creativity Support Index	I was satisfied with what I got out of [System].	0–10
CSI6	Creativity Support Index	What I was able to produce was worth the effort I had to exert to produce it.	0–10
SR1	System Rating	[System] helped me clarify an initial visual direction for the scientific figure before I committed to a detailed design.	1–7
SR2	System Rating	[System] helped me turn the scientific content into an inspectable structure of components, relationships, data flow, and reading order.	1–7
SR3	System Rating	[System] allowed me to explore alternative figure candidates and choose which direction to develop further.	1–7
US1	Overall System Usability	[System]’s capabilities meet my requirements.	1–7
US2	Overall System Usability	[System] is easy to use.	1–7
SO1	Sense of Ownership	To what extent do you feel that the final figure you produced is yours?	1–7
SO2	Sense of Ownership	To what extent does the final scientific figure reflect your own understanding of the assigned method and source material?	1–7
UA1	User Agency	How much control did you feel over the process of organizing, understanding, and visually synthesizing the source material?	1–7
UA2	User Agency	How much control did you feel over the final figure you produced?	1–7
WL1	Mental Effort	How much mental effort did you invest in completing the current task with [System]?	1–9
WL2	Task Difficulty	How difficult did you find this scientific-figure synthesis task?	1–9

Notes. For the Creativity Support Index items, 0 = Highly disagree and 10 = Highly agree. For the System Rating, Overall System Usability, Sense of Ownership, and User Agency items, 1 = Highly disagree and 7 = Highly agree. For the Mental Effort and Task Difficulty items, 1 = Very, very low and 9 = Very, very high. [System] was replaced with *GPT Baseline* or *HiFigure* in the corresponding questionnaire.

End-to-end Method. Instruction-aware Scene Memory identifies objects and uncertain regions. The Active View Controller acquires the feasible view with maximum expected information gain, and the Evidence Integrator updates memory. The Safe Task Policy executes an action only after the safety check passes.

Completion Requirements.

- (1) Maintain object, temporal, and uncertainty state from language and vision.
- (2) Choose the feasible view with maximum expected information gain under the sensing budget.
- (3) Align the refreshed observation and update memory confidence.
- (4) Execute the task action only after its safety check passes.

Participant Instruction. Please use the assigned system to generate a scientific figure that satisfies the requirements above. Take a screenshot of the selected figure or copy it into the designated area.

C Post-task Questionnaires

After completing each task, participants completed a system-specific questionnaire. The Baseline questionnaire was administered after using GPT Baseline, and the HiFigure questionnaire was administered after using HiFigure. Both questionnaires contained the same constructs, items, and response scales, with only the system name changed. In the tables below, [System] refers to either *GPT Baseline* or *HiFigure*.

Table 2. Items used to evaluate the generated scientific figures in GPT Baseline and HiFigure System.

ID	Construct	Item	Scale
<i>PREVis: Perceived Readability</i>			
PR1	Perceived Readability	It is obvious to me how to read this visualization.	1-7
PR2	Perceived Readability	I can easily understand how the data are represented in this visualization.	1-7
PR3	Perceived Readability	I can easily understand this visualization.	1-7
PR4	Perceived Readability	I do not find this visualization messy.	1-7
PR5	Perceived Readability	I do not find this visualization crowded.	1-7
PR6	Perceived Readability	I do not find distracting parts in this visualization.	1-7
PR7	Perceived Readability	I can easily find specific elements in this visualization.	1-7
PR8	Perceived Readability	I can easily identify relevant information in this visualization.	1-7
PR9	Perceived Readability	I can easily retrieve information from this visualization.	1-7
<i>Overall Figure Rating</i>			
FR1	Usefulness	To what extent is this system-generated figure useful for communicating the method's core components, relationships, and main contribution?	1-7
FR2	Beauty	To what extent is this figure visually appealing?	1-7
FR3	Faithfulness	To what extent does this figure faithfully and accurately represent the assigned method and source material?	1-7
FR4	Completeness	To what extent does this figure include the essential inputs, components, stages, outputs, and relationships needed to understand the assigned method?	1-7
FR5	Correctness	To what extent do the components, arrows, and dependencies in this figure form a logically correct and coherent structure?	1-7

Notes. All items were rated from 1 (*Strongly disagree*) to 7 (*Strongly agree*). PR1-PR9 are PREVis perceived-readability items; FR1-FR5 assess the overall quality of the generated figure.

D Post-task Semi-Structure Interview Outline

After completing the tasks with both systems, participants took part in a 6-8 minute semi-structured interview. The interview focused on participants' prior experience with AI-assisted research figure design and their comparative experiences with HiFIGURE and the baseline system. The interview questions were as follows:

- (1) **Background and prior practice.** What is your research area? How do you usually collaborate with AI to design research figures? What platforms and models do you primarily use?
Optional probe: What did the system contribute?
- (2) **Interaction with AI.** Which system do you think can better help you interact with AI? Why?
Optional probe: Can you point to one specific example?
- (3) **Exploration of alternatives.** Which system do you think is better suited for exploring different solutions?
Optional probe: Were the alternatives meaningfully different?
- (4) **Agency and contribution.** When using each system, when did you feel in control of the process, and when did you feel restricted? Which parts of the final image did you feel were your contribution?
Optional probe: What caused that feeling?
- (5) **Preference and future expectations.** Which system would you choose to use? What would your ideal system be?
Optional probe: Why would that capability matter?

E Detailed Expert Evaluation Rubric

Experts evaluated each generated figure on eight dimensions according to FigureBench [48] with a five-point scale, where 1 indicates *Very Weak*, 2 *Weak*, 3 *Adequate*, 4 *Strong*, and 5 *Excellent*. Table 3 summarizes the evaluation criteria and score anchors.

Table 3. Expert evaluation rubric for generated scientific figures.

Dimension	Definition	1: Very Weak	2: Weak	3: Adequate	4: Strong	5: Excellent
Aesthetic & Design Quality	Visual appeal, composition, layout, and design innovation.	Rough or disordered. Visual choices seriously impede reading.	Basic and poorly coordinated, with several layout or design problems.	Clear and usable, but conventional or somewhat stiff.	Modern and cohesive, with intentional composition and design detail.	Highly unified and visually strong. Design reinforces the scientific message.
Visual Expressiveness	Use of meaningful graphics, icons, and visual metaphors to convey abstract concepts.	Little effective visual encoding, or graphics conflict with their meaning.	Some useful graphics, but most are decorative or minimally expressive.	Key concepts are understandable, though the visual language is conventional.	Graphics and encodings have clear semantic roles and aid understanding.	Rich, precise, and original visual language conveys concepts with little reliance on text.
Professional Polish	Quality of alignment, spacing, scale, typography, color, and stylistic consistency.	Severe misalignment, overlap, clipping, illegible text, or unfinished elements.	Multiple inconsistent or untidy details create a draft-like appearance.	Generally finished, with a few visible alignment, spacing, or styling issues.	Careful and consistent execution with only minor flaws.	Publication-grade finish with precise and consistent visual details.
Clarity	Legibility and accessibility of the main message, hierarchy, and details.	Main message is difficult to identify and the figure is nearly incomprehensible.	Some meaning is visible, but poor hierarchy or legibility requires substantial guessing.	Main information is understandable, though local density or hierarchy slows reading.	Message and details are clear, with hierarchy supporting efficient reading.	Complex information is immediately clear and every visual choice supports understanding.
Logical Flow	How sequence, arrows, grouping, and dependencies guide the reader.	Flow is missing or contradictory, with no comprehensible path.	A rough sequence exists, but connections are ambiguous or inconsistent.	Main flow is traceable, with some unclear branches, feedback, or dependencies.	Sequence, grouping, and connections clearly guide the reader.	Complex branches, feedback, and dependencies are rigorous and unambiguous.
Accuracy	Faithfulness to the assigned components, relationships, and method constraints.	Core method is misrepresented with multiple misleading relationships or statements.	Some core content is correct, but notable errors or unsupported elements remain.	Main content is broadly accurate, with minor simplifications or ambiguities.	Nearly all components and relationships are faithful and accurate.	Representation is precise and traceable, with no unsupported or misleading content.
Completeness	Coverage of key inputs, components, stages, outputs, and relationships.	Most critical content is missing and the method cannot be reconstructed.	Some main content is present, but critical stages, outputs, or relationships are missing.	Most necessary content is covered, with a few secondary omissions.	Key inputs, components, stages, outputs, and relationships are essentially complete.	All necessary information is present without critical omissions or irrelevant excess.
Appropriateness	Fit of complexity, abstraction, terminology, and visual style for the target paper audience.	Presentation is severely mismatched to the intended audience or setting.	Weak audience fit due to unsuitable complexity, terminology, or style.	Generally suitable, with some aspects of abstraction or density needing adjustment.	Complexity, terminology, and style are well matched to the target audience.	Exceptionally well calibrated to the publication context, balancing professionalism and readability.

F User Authoring Results

Table 4. Descriptive statistics and paired comparisons for system-level evaluation ($N = 30$).

Measure	Baseline M (SD)	HiFIGURE M (SD)	ΔM	t (29)	p_{Holm}	d_z
Creativity Support Index	5.58 (2.02)	8.31 (1.34)	2.73	6.18	2.94×10^{-6}	1.13
Workload	4.40 (1.48)	4.13 (1.96)	-0.27	-0.55	0.588	-0.10
System Rating	4.20 (1.14)	6.14 (0.71)	1.94	8.60	1.08×10^{-8}	1.57
System Usability	4.63 (1.50)	6.00 (0.83)	1.37	4.40	2.68×10^{-4}	0.80
Sense of Ownership	4.12 (1.11)	5.98 (0.68)	1.87	8.26	2.10×10^{-8}	1.51
User Agency	3.87 (1.14)	5.82 (1.02)	1.95	7.15	2.86×10^{-7}	1.31

Note. Creativity Support Index scores range from 0 to 10; Workload scores range from 1 to 9; all other scores range from 1 to 7. Workload is the mean of mental effort and task difficulty, with lower scores indicating lower perceived workload. Higher scores indicate more positive evaluations for all other measures. Multi-item scores are arithmetic means. $\Delta M = M_{\text{HiFigure}} - M_{\text{Baseline}}$. All six comparisons used two-sided paired t -tests ($df = 29$), following Shapiro-Wilk tests of the paired differences. Holm correction covers the six system-level comparisons. d_z is the mean paired difference divided by its sample standard deviation; its sign follows the HiFigure-minus-Baseline contrast.

G User Figure Results

Table 5. Descriptive statistics and paired comparisons for user-rated figure evaluation ($N = 30$).

Measure	Baseline M (SD)	HiFIGURE M (SD)	ΔM	Statistic	p_{Holm}	d_z
<i>Perceived Readability (PREVis)</i>						
Understandability	4.43 (1.11)	5.96 (0.73)	1.52	$t = 7.27$	3.69×10^{-7}	1.33
Layout Clarity	3.87 (1.30)	5.81 (0.96)	1.94	$t = 7.23$	3.69×10^{-7}	1.32
Reading Data	4.18 (1.09)	6.04 (0.73)	1.87	$t = 7.55$	2.05×10^{-7}	1.38
Overall Figure Rating	4.26 (1.00)	6.01 (0.60)	1.75	$t = 9.77$	1.02×10^{-9}	1.78
Usefulness	4.13 (1.07)	6.20 (0.66)	2.07	$W = 0$	7.73×10^{-6}	2.04
Beauty	4.23 (1.07)	6.30 (0.84)	2.07	$W = 2$	1.20×10^{-5}	1.76
Faithfulness	4.03 (1.40)	5.87 (0.86)	1.83	$t = 6.55$	1.78×10^{-6}	1.20
Completeness	4.80 (0.92)	5.87 (0.78)	1.07	$W = 24.5$	2.41×10^{-4}	0.93
Correctness	4.10 (1.30)	5.80 (1.10)	1.70	$t = 5.67$	1.20×10^{-5}	1.03

Note. All scores range from 1 to 7, with higher scores indicating more positive evaluations. Each PREVis subscale is the mean of three items; Overall Figure Rating is the mean of the five task-specific items listed below it. $\Delta M = M_{\text{HiFigure}} - M_{\text{Baseline}}$. Statistics are from two-sided paired t -tests ($df = 29$) or Wilcoxon signed-rank tests (W), selected using the Shapiro-Wilk test on paired differences. Holm correction covers all nine comparisons. For every measure, d_z is the mean paired difference divided by its sample standard deviation.

H Detailed Expert Evaluation of Figure Quality

Table 6 reports the individual dimension scores for both experts. Each comparison uses 30 participant pairs. Holm correction was applied across the eight dimensions and the overall score separately for each expert. Positive d_z values indicate higher scores for HiFIGURE.

Table 6. Expert evaluations of the eight figure-quality dimensions.

Dimension	Baseline $M(SD)$	HiFIGURE $M(SD)$	p_{Holm}	d_z
Expert 1				
Aesthetic and Design Quality	2.17 (1.02)	3.17 (0.99)	< .001	1.02
Visual Expressiveness	2.43 (0.94)	3.50 (0.86)	< .001	1.13
Professional Polish	2.70 (0.75)	3.50 (0.78)	.001	0.87
Clarity	2.77 (0.82)	3.53 (0.73)	.002	0.76
Logical Flow	2.43 (0.57)	3.17 (0.65)	.001	0.93
Accuracy	2.83 (0.87)	3.60 (0.67)	.001	0.89
Completeness	2.83 (0.87)	3.60 (0.67)	.002	0.79
Appropriateness	2.63 (0.81)	3.57 (0.68)	< .001	1.03
Expert 2				
Aesthetic and Design Quality	2.53 (0.82)	3.83 (0.83)	< .001	1.27
Visual Expressiveness	2.37 (1.07)	3.80 (0.81)	< .001	1.12
Professional Polish	2.47 (0.82)	3.63 (0.72)	< .001	1.47
Clarity	2.17 (0.87)	3.87 (0.97)	< .001	1.61
Logical Flow	2.40 (0.81)	3.87 (0.73)	< .001	1.57
Accuracy	2.83 (0.79)	3.67 (0.71)	< .001	0.88
Completeness	2.57 (0.82)	3.53 (0.82)	< .001	0.97
Appropriateness	2.13 (0.82)	3.67 (0.80)	< .001	1.47

I Prompts

This section presents condensed versions of the main prompts referenced in the backend description. We retain the system–user separation and the input variables used by the implementation, while omitting output schemas, retry instructions, and other implementation-specific constraints.

The *Retrieval Query Decomposition Prompt* is used by *Retrieval* to transform the current figure description into separate Topic, Layout, and Style queries for the corresponding retrieval indices.

System Prompt

Prepare retrieval queries for scientific figure references from a single drawing prompt.

Topic (idea_query): preserve the research problem, core mechanism, inputs and outputs, and explanatory intent.

Layout (structure_query): emphasize modules, topology, reading order, branches, loops, and connections. Do not invent unspecified structure.

Style (style_query): include only explicitly requested visual language, palette, typography, and rendering preferences.

User Prompt

<PROMPT>

The *Reference-Conditioned Layout Planning Prompt* is used by *Layout* to derive the scientific structure, abstract the selected layout reference, and map the structure onto the reference-derived organization. The prompt below condenses the three corresponding internal planning calls into their shared task.

System Prompt

Extract the major scientific components and directed relationships from the brief without adding visual style. Abstract the layout reference into a neutral framework that preserves panel structure, relative geometry, whitespace, branching, and reading flow. Map every scientific component to this framework while preserving semantic dependencies and assigning each component exactly once.

User Prompt**USER BRIEF**

<USER_BRIEF>

LAYOUT REFERENCE

<LAYOUT_REFERENCE>

The *Candidate Generation Prompt* is used by *Candidate* to compile the current scientific content and visual constraints into the instruction sent to the image-generation backend. In the implementation, this generation request is sent as a single user-role prompt rather than a separate system-user pair.

System Prompt

Not separately used. The compiled generation instruction is sent as a user-role input to the image-generation model.

User Prompt**TASK**

Generate one clean, publication-quality scientific figure.

AUTHORITY RULES

The <USER_GOAL> and figure structure determine scientific content, labels, and topology. The <LAYOUT_REFERENCE> guides macro composition and reading order. The <LAYOUT_SKELETON> preserves modules and relationships. The <STYLE_REFERENCE> guides palette, typography, shapes, strokes, spacing, and rendering. The <MATCH_CONSTRAINTS> override only their explicitly specified targets.

USER GOAL

<USER_GOAL>

REFERENCES

<REFERENCE_MANIFEST>

STYLE

<STYLE_CONTRACT>

MATCH

<MATCH_CONSTRAINTS>

OUTPUT

Return one standalone figure. Preserve the specified scientific modules and relationships, and do not introduce, remove, merge, or rename them.